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ACADEMIC INTERNSHIP REPORT

DESIGN AND IMPLEMENTATION OF A CROSS PLATFORM INTEGRATED MANAGEMENT APPLICATION FOR HAJJ

Internship carried out from the 1st July to 30th September 2026 in view of obtaining an
Engineering Diploma in Computer Sciences

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Academic Year 2025 - 2026

DEDICATION

**TO THE
MAMOUDOU's
FAMILY**

ACKNOWLEDGEMENTS

- My gratitude goes to the following people who assisted in the realization of this project:
- The Resident Representative of AICS-Cameroon, **Mr. Armand Claude ABANDA**, for his words of encouragement and the different advices brought to us.
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GLOSSARY

- ❖ 2TUP: Two Track Unified Process.
- ❖ AICS: African Institute of Computer Sciences
- ❖ MVC: Model View Controller.
- ❖ UML: Unified Modelling Language
- ❖ DBMS: Database Management System
- ❖ GUI: Graphical User Interface
- ❖ HAJJ: Muslim pilgrimage
- ❖ HNC: Hajj National Commission

ABSTRACT

In the context of digital transport of public service, the organization of the Hajj in Cameroon remains largely manual, resulting in inefficiencies, delays, and poor coordination. This thesis proposed the design and development of an integrated mobile application aimed at improving the overall Hajj experience for Cameroonian pilgrims .The application allows for secure online registration, document submission, real-time updates ,logistical tracking, and post-pilgrimage follow-up Through a user- centre design methodologies, a functional prototype has been developed .That is why we came up with the topic “DESIGN AND DEVELOPMENT OF A CROSS-PLATFORM INTEGRATED MANAGEMENT APPLICATION FOR HAJJ IN CAMEROON” This project represents a significant step toward digital innovation in religious public services aims to enhance the organization and security of pilgrimage management in Cameroon

Key Words:

- Hajj management
- Cameroonian pilgrims
- Digital transformation

RESUME

Dans le contexte de la transformation numérique des services publics, l'organisation du Hajj au Cameroun reste en grande partie manuelle, ce qui entraîne des inefficacités, des retards et une peu bonne coordination. Ce mémoire propose la conception et le développement d'une application mobile intégrée visant à améliorer l'expérience globale du Hajj pour les pèlerins Camerounais .L'application permet une inscription en ligne sécurisée ,la soumission de documents ,des mises à jour en temps réel, le suivie logistique ,ainsi qu'un accompagnement post-pèlerinage .C'est pourquoi nous avons proposé ce sujet<<CONCEPTION ET DEVELOPPEMENT D'UNE APPLICATION CROSS PLATFORM DE GESTION INTEGREE DU HAJJ AU CAMEROON Grâce à une méthodologie de conception centrée sur l'utilisateur et à l'utilisation des technologies mobiles modernes, un prototype fonctionnel a été développé .Ce projet représente une avancée significative vers l'innovation numérique dans les services publics religieux et la sécurité de la gestion du pèlerinage au Cameroun

Mots Clés :

- Gestion du Hajj
- Pèlerins Camerounais
- Transformation numérique

GENERAL INTRODUCTION

Hajj, the fifth pillar of Islam, is a major religious obligation for every Muslim who possesses the physical and financial means to undertake it. Each year, thousands of Cameroon pilgrims aspire to fulfil this sacred duty under the best possible conditions. However, organization of the Hajj in Cameroon is often marked by numerous challenges: slow and centralized registration processes, lack of adequate digital tools suited to the local context, etc. Second year students in AICS Cameroon, are required to carry out a three (3) months academic internship at an enterprise to put into practice the knowledge gotten from school in the professional milieu. For this reason, we applied for an internship at SHADERL, which specializes in providing software solutions while there, we were attributed the theme” **DEVELOPMENT AND DESIGN OF A CROSS-PLATFORM INTEGRATED MANAGEMENT SYSTEM FOR HAJJ IN CAMEROON**” We divided this report several parts which are as follows:

1. Insertion phase
2. Existing system
3. Specification book
4. Analysis phase
5. The Conception phase
6. The Deployment phase or Realization
7. Functionality or Testing phase
8. The User Guide

PART I : INSERTION PHASE

Preamble

This section of the report will cover details of how we were welcomed in the host company, presentation, organization and brief introduction to our project.

Content

INTRODUCTION

- I. WELCOME AND INSERTION
 - II. GENERAL PRESENTATION OF THE COMPANY
 - III. ORGANIZATION OF THE COMPANY
 - IV. GEOGRAPHICAL LOCATION OF SHADERL
 - V. BRIEF PRESENTATION OF PROJECT THEME
- ### CONCLUSION



INTRODUCTION

The insertion phase in a company is a period during which we discover our working environment, the staff of the company and other interns. Here, we will begin by discussing our first two weeks in the company, how we were welcomed and how we began adapting to our internship environment, we will proceed by exploring the history of the company, discovering its missions, learning what its major activities are, and witnessing some of its key realizations. We will also get to understand how the company is structured administratively and functionally so that it operates effectively and accomplishes its goals. Furthermore, we will look at the hardware equipment used by the company, and the software resources used in its daily operations. We will then introduce our chosen theme for the internship period, briefly elaborate on It.

I-WELCOME AND INSERTION

We arrived at **SHADERL** on Wednesday 01st July 2026, at 08:30 am. we were received by the SHADERL team who introduced us to our workspace, gave us an official welcome to the enterprise, its activities and its different rules and regulations. Also, we discussed on the enterprise's preferred languages and frameworks, our professional supervisor encouraged us through his past working experiences in different enterprises.

The first two week we started with our internship report by configuring our MS WORD processor, reviewing key concepts like page numbering, styles, section breaks, page breaks etc. There was a talk on project ideas, we were advised to propose project ideas and we were assigned the tasks to carry out research on these ideas. The objective was that the enterprise did not want to impose some themes on us. For those who could not find a theme, the enterprise could propose an idea for them to think about.

II-GENERAL REPRESENTATION OF SHADERL

A-Background:

SHADERL is a dynamic, non-governmental tech startup founded in 2023 by **Mr. Yvan EMATE** with a bold vision: to bridge the gap between traditional industries and the rapidly evolving digital economy. Recognizing the urgent need for digital transformation, SHADERL is committed to providing **cutting-edge IT solutions** while empowering the next generation of innovators.

B- Mission

At SHADERL, we harness the power of **AI and cutting-edge technology** to democratize digital transformation. Our mission is to **empower individuals and businesses** with intelligent tools, accessible education, and personalized mentorship bridging the gap between today's potential and tomorrow's opportunities. We believe AI should **augment human potential, not replace it**. By making AI-driven solutions practical and ethical, we equip the next generation to lead, innovate, and thrive in an era where **human creativity meets machine intelligence**.

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C-Vision

At SHADERL, we envision a world where **technology unites humanity**, where innovation is driven by collective purpose, not just profit. By putting “**PeopleFirst**”, we strive to build a future where:

-**One Love**→ Technology fosters **inclusion, empathy, and global collaboration**, breaking down barriers.

-**One Mind**→ AI and digital tools **amplify human potential**, creating shared knowledge and opportunity.

-**One Legacy**→ Every individual we empower leaves a lasting impact, shaping a **smarter, kinder, and more connected world**.

We don't just adapt to the digital age; we **redefine it with humanity at the core**.

D- ACTIVITIES

The activities of SHADERL range from computer sciences, engineering, and training. We can outline the following:

- Conception realization, and hosting of websites.
- Software development and maintenance.
- Training in Software related fields.
- Conception and realization of multimedia.
- AI driven solutions
- IT consulting and innovation.

E- PRODUCTS

NO	Name	description	LOGO
1	ShaderlStock	ShaderlStock is an application, which helps small, medium and large businesses to manage their inventory, with the help of a QR code and AI assistance for more complex queries.	 
2	ShaderlAlgorithm	It is an application, which helps individuals in the society to learn how to write algorithms and programing, despite their educational background. With the help of artificial intelligence.	

III-ORGANISATION OF SHADERL

SHADERL is administratively organized as follows:

a- Executive Leadership

This department is responsible for:

- ✓ Setting the company's vision, mission, and strategic direction
- ✓ Making high-level decisions about investments, partnerships, and growth
- ✓ Representing SHADERL in key meetings with investors and government agencies
- ✓ Ensuring all departments align with the company's core values and objectives
- ✓ Overseeing the overall performance and sustainability of the organization

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b. Software Engineering Department

This department is responsible for:

- ✓ Designing, developing, and maintaining all of SHADERL'S software products
- ✓ Implementing AI and machine learning solutions for company offerings
- ✓ Ensuring software security, scalability, and optimal performance
- ✓ Collaborating with other departments to understand technical requirements
- ✓ Staying updated with emerging technologies and industry best practices

c. Human Resource Department

This department is responsible for:

- ✓ Recruiting and onboarding top talent that aligns with SHADERL'S values
- ✓ Managing employee relations, welfare, and performance evaluations
- ✓ Developing training programs to enhance staff skills and capabilities
- ✓ Maintaining company culture and ensuring a positive work environment
- ✓ Handling compensation, benefits, and conflict resolution

d. Communication Department

This department is responsible for:

- ✓ Managing SHADERL'S public image and brand reputation
- ✓ Developing and executing marketing and PR strategies
- ✓ Handling all internal and external communications
- ✓ Managing social media platforms and digital content
- ✓ Organizing corporate events and press engagements

e. Department of Financial Affairs

This department is responsible for:

- ✓Managing all financial operations and accounting
- ✓Preparing budgets and financial forecasts
- ✓Handling payroll, taxes, and financial reporting
- ✓Ensuring compliance with financial regulations
- ✓ Managing investments and financial partnerships

B- Functional organisation

The functional branch of Shaderl is organized as follows:

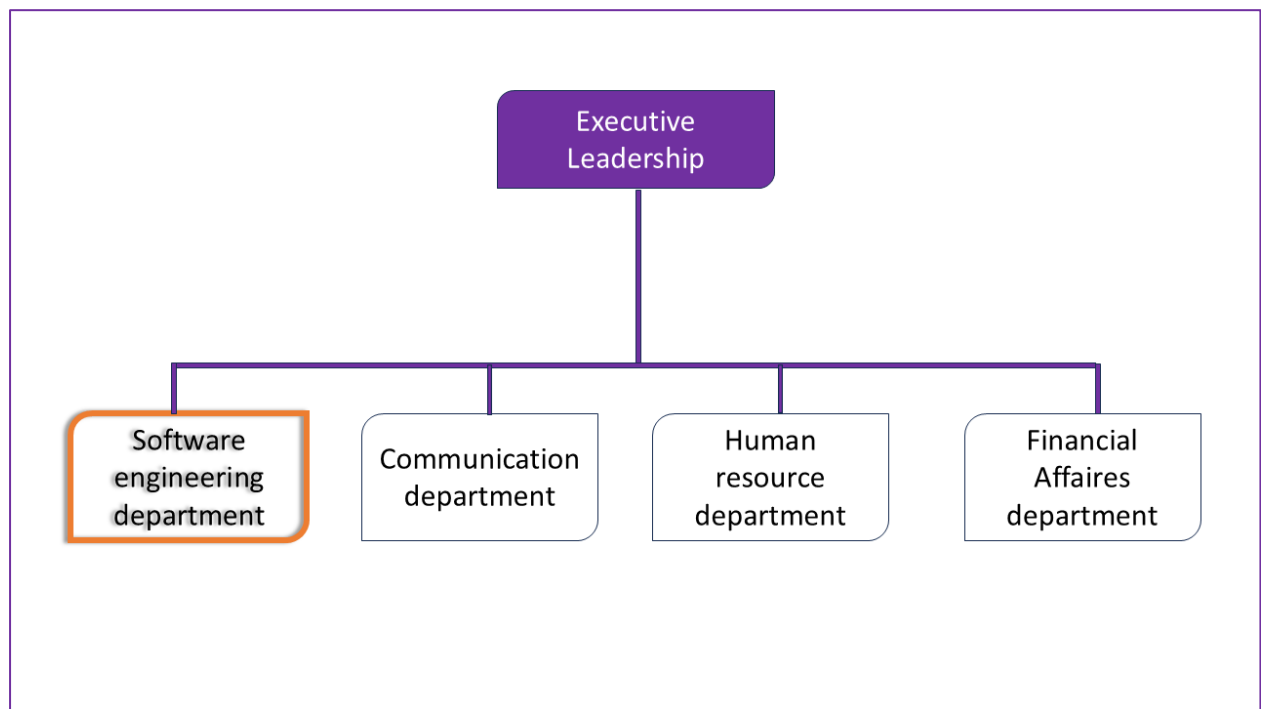


Figure 1: Fonctionnal organisation of SHADERL

IV- GEOGRAPHICAL LOCATION OF SHADERL

Coming from AWAE junction, the location of SHADERL is facilitated by the map below.

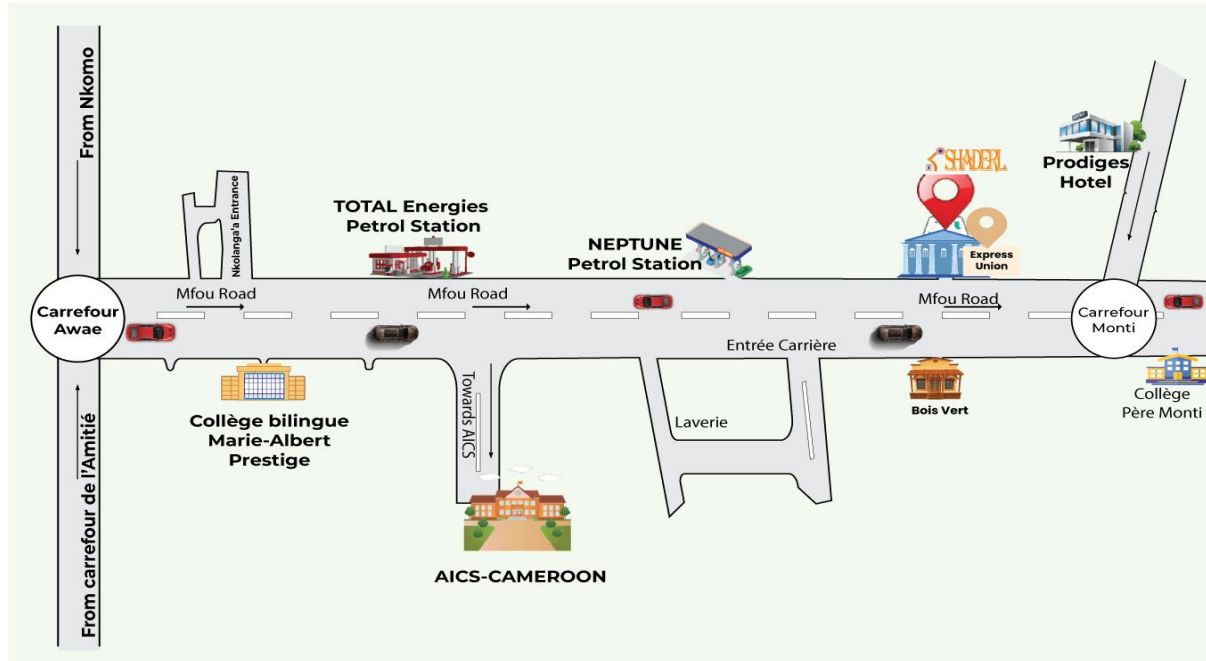


Figure 2: Geographical location of SHADERL

V- BRIEF PRESENTATION OF THE PROJECT THEME

Following our arrival at SHADERL, we were encouraged to suggest project ideas for our internship. Following the mission of SHADERL to take active part in the development of software applications, we saw an environment for creativity in our project idea and proposed the theme “DEVELOPMENT AND DESIGN OF A CROSS-PLATFORM INTEGRATED MANAGEMENT APPLICATION FOR THE HAJJ IN CAMEROON “This system aims to provide a centralised digital solution that ensures transparency, efficiency, accessibility for all stakeholders involved: administrative bodies, medical approval, and pilgrims themselves. The system will be accessible across platforms (mobile and web), allowing users to register, track their progress, receive updates, manage documentation-all in one place. This theme is both technically and society impactful, as it aligns with national efforts to digitalize public services while offering me the opportunity to apply key concepts like;

- ❖ Software architecture
- ❖ Cross-platform mobile development
- ❖ Design
- ❖ Management
- ❖ API integration

PART I : TECHNICAL PHASE

Preamble

The technical phase is a document that describes the process, progress, or results of a scientific research. It also includes a detailed analysis and the conclusions drawn from that research.

BOOK I: THE EXISTING SYSTEM

BOOK II: THE SPECIFICATION BOOK

BOOOK III: THE ANALYSIS PHASE

BOOK IV: CONCEPTION PHASE

BOOK V: REALIZATION PHASE

BOOK VI: TEST OF FUNCTIONALITIES

BOOK VII: USER GUIDE

BOOK I : THE EXISTING SYSTEM

Preamble

The existing system is a document that provides a view of how the current system works, how it carries out its different activities, it equally provides a deep understanding of the system associated to the various limitations and the problems that result from it and we propose the solution

Content

INTRODUCTION

- I. PRESENTATION OF THE PROJECT THEME
- II. STUDY OF THE EXISTING SYSTEM
- III. CRITICISMS OF THE EXISTING SYSTEM
- IV. PROBLEMATICS
- V. PROPOSED SOLUTION

CONCLUSION

I. Presentation of the Project Theme

1.1. Context

Hajj is a significant religious obligation for Muslims that involves complex logistics and coordination. In Cameroon, managing the Hajj process—covering registration, visa issuance, health checks, travel arrangements, accommodation, and post-pilgrimage follow-up—remains largely manual or semi-automated, with disparate systems and inefficiencies.

1.2. Objective

The project aims to **Design and develop a cross-platform integrated application** that supports the **end-to-end management** of Hajj operations in Cameroon, from initial pilgrim registration through to post-Hajj feedback and monitoring.

II. Study of the Existing System

2.1. Current Process Overview

a) . Registration And Coordination in Cameroon

Cameroon National Hajj Commission (HNC)

It is the body in charge of coordinating Hajj activities in Cameroon it is under the ministry of territorial administration of Cameroon and its president is Mr ATANGA NJI Paul

- All Hajj aspirants must register through the HNC, the official body overseeing Hajj from Cameroon.[minat.gov.cm+4minat.gov.cm+4Hajj Reporters+4](#)
- In recent years, Cameroon has signed bilateral agreements with Saudi Arabia and service providers (e.g., Rawaf Mina Company) to coordinate logistics, quotas, transport, accommodations, and pilgrims' welfare.[minat.gov.cm](#)

b. Registration Process

- Starting months before Hajj, registration opens via a **single official window**, often through banks (e.g. Afriland First Bank). For example, in 2025 the deadline was around mid-March.[Mimi Mefo Info](#)
- Pilgrims submit required documents: valid passport, photographs, vaccination certificates, proof of payment, and health clearance.[AWAM PK+1ambacamsaudi.org+1](#)
- Payment covers package cost (FCFA 3 million+ in 2025) and registration fee, though government subsidies help reduce burden.[Mimi Mefo Info+1minat.gov.cm+1](#)
- After reaching quota (e.g. ~4,527 slots for Cameroon in 2025), registration closes early.[qurantime.netHajj Reporters](#)

c. Verification & Package Selection (Saudi Nusuk)

- Package selection and verification are completed via the Saudi government's **Nusuk platform** integrated with **Tasreeh/ Tawakkalna**.[Hajj Reporters+14AWAM PK+14Reddit+14](#)
- Applicants choose their preferred “Hajj package” (includes flights, accommodation, meals, guide). Timeline: packages preview from ~20 Jan; purchase begins early Feb.[Reddit+1Reddit+1](#)

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- Verification of applications typically takes from hours to a couple of weeks, depending on system load and manual review.[RedditRedditReddit](#)

2. Pre-departure Logistics and Departure from Cameroon Travel Arrangements

- Cameroon's HNC coordinates flights—typically through **Ethiopian Airlines**, handling around 4,000 pilgrims each season.[minat.gov.cm+1minat.gov.cm+1](#)

Subsidy and Final Briefing

- Before departure, pilgrims may receive government subsidies (e.g. FCFA 350,000 each in 2025).[minat.gov.cm+1Mimi Mefo Info+1](#)
- Pilgrims gather at **Yaoundé-Nsimalen International Airport Douala International Airport and Garoua International Airport** for check-in. Standard departure procedures are followed: biometric screening, visa checks, security controls.

3. Arrival in Saudi Arabia and During Hajj

Arrival and Entry Clearance

- Under the “**Road to Makkah**” initiative, some countries permit **immigration clearance in Cameroon or Addis Ababa** to speed entry—though Cameroon isn't formally listed, coordination may offer simplified transit.[minat.gov.cm](#)
- Pilgrims land in Saudi Arabia, pass through immigration, collect baggage, and join group transport to accommodation in Mecca or Medina.

E-Track and Services Monitoring

- Saudi's **E-Track system** tracks the delivery of contracted services (transport, housing, catering) from arrival to departure.[MyGov](#)

ID and Identification

- Each pilgrim receives a **bracelet and/or ID card** encoding personal and group data—required for access to holy sites and crowd control.[Wikipedia](#)

Health, Safety and Conduct

- Required vaccines: meningitis ACYW135 (valid up to 3 years), yellow fever (if applicable), flu. COVID requirements vary year-to-year. [AWAM PK](#)
- Pilgrims must remain with their group, avoid unauthorized travel outside permitted zones, keep personal documents secure, and follow local guidance issued by Cameroon Embassy or Consulate. [ambacamsaudi.orgWikipedia](#)

Performing Rituals

- Pilgrims move through Mecca's rituals (Tawaf, Sa'i, staying in Mina, Arafat, Muzdalifah, stoning Jamarat), using designated transport and, during peak days, the Al-Mashaaer Metro where available. [Wikipedia](#)

4. Return Planning and Departure to Cameroon

Ending the Hajj

- All pilgrims must **exit Saudi Arabia before the 10th of Muharram** (the first month following Dhu'l-Hijjah). Overstays can lead to fines ($\approx 10,000$ SAR) or immigration penalties. [ambacamsaudi.org](#)

Transport to Airport and Departure

- Pilgrims reconvene in their groups at scheduled departure dates (often coordinated by HNC and travel agent).
- Transportation to major airports (e.g. Jeddah, Medina) is organized; flight is typically return via Ethiopian Airlines.

Arrival in Cameroon

- Upon arrival in Yaoundé, Garoua or Douala, pilgrims go through immigration and customs. The national Hajj authority or airport staff may assist group reunification and wellness checks.

Quick Timeline Example (Based on 2026 cycle)

Table 1: quick timeline for 2025 source; wikipedia

Step	Approx. Date	Event
Registration opens	Dec 2025 – Mar 2026	Submit via HNC
Nusuk Package preview	~Jan 20, 2026	Review packages
Purchase begins	Early Feb 2026	Secure package
Verification results	Within days–weeks	Via Nusuk
Departure flight	~May 8, 2026	From Yaoundé
Hajj rituals	~Late May–	Dhu'l-Hijjah (~24 May–n 2026) Wikipedia +1 Deutsche Welle +1 minat.gov.cm Reddit +7 Reddit +7 Reddit +7 ambacamsaudi.org +6 Reddit +6 minat.gov.cm +6
Return home	Before 10 Muharram (~early July)	Finish pilgrimage

III. Criticisms of the Existing System

- **Lack of Integration:** No centralized platform to manage pilgrim data across ministries.
- **Data Redundancy & Inconsistency:** Multiple registrations often result in duplicate or conflicting information.
- **No Real-Time Monitoring:** Authorities lack visibility into the status and location of pilgrims.
- **Poor User Experience:** Paper-based processes are time-consuming and error-prone.
- **Limited Accessibility:** Systems are not accessible across devices (e.g., mobile/tablet/web).
- **Lack of Feedback Loop:** No mechanism to collect, analyze, and act on post-Hajj data from pilgrims.

IV. Problematic

How can the Hajj management process in Cameroon be streamlined and modernized to ensure transparency, efficiency, accessibility, and effective post-Hajj monitoring through a single, integrated, and cross-platform digital solution?

V. Proposed Solutions

To address system shortcomings, such as verification delays, server overload during Nusuk package selection, fund, and pilgrim welfare gaps—it is recommended to integrate our national Hajj platform with a **centralized digital coordination layer** that provides real-time tracking, status notifications, GPS-based transport monitoring, automated alerts, and a streamlined complaint-resolution interface. This would mitigate bottlenecks experienced on Nusuk, ensure transparency in payment flow, reduce fraud by parsing guidance and verification through accredited channels, and link with Cameroon's one-stop registration model. Complementing this, establish enhanced **operational oversight protocols** including timely remittance and budget audit controls, standardized welfare standards (trained motawifs, medical accompaniment, decent accommodations), and proactive safety infrastructure (e.g., heat-mitigation, signage, crowd control in Mina/Arafat). Together, these reforms embedded in your existing system would strengthen efficiency, accountability, and pilgrim welfare throughout the Hajj cycle. [reddit.com+7reddit.com+7reddit.com+7](#)

5.1. Core Solution

Develop a **cross-platform integrated Hajj management application** accessible via **web and mobile**, covering:

- **Pilgrim Registration Module** (Biometric, Document Upload, Payment Integration)
- **Visa Application Integration** (with Saudi Embassy systems)
- **Health Check and Vaccination Monitoring**
- **Travel and Logistics Coordination** (flight scheduling, hotel bookings)
- **Pilgrim Tracking System** (real-time location monitoring during Hajj)
- **Post-Hajj Feedback and Monitoring Dashboard**
- **Role-Based Access** for MINREX, MINAT, medical personnel, and travel agencies
- **Data Analytics and Reporting Tools**

5.2. Technologies to Use

- **Frontend:** Flutter or React Native (for cross-platform compatibility)

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- **Backend:** Node.js / Laravel
- **Database:** MySQL
- **Authentication:** OAuth 2.0, Biometric support
- **APIs:** RESTful APIs
- **Cloud Deployment:** AWS / Azure / Google Cloud

5.3. Expected Benefits

- **Reduced Processing Time** for registration and visa application
- **Integrated Workflow** across all stakeholders
- **Accessible Anywhere** on web for pilgrim guide and mobile platforms for pilgrims
- **Secure & Scalable Infrastructure**
- **Better Decision-Making** through real-time analytics
- **Improved Communication** between agencies and pilgrims
- **Feedback-Driven Improvements** to the Hajj process annually

VI. Conclusion

The Hajj pilgrimage is a deeply spiritual journey that deserves smooth and dignified facilitation. By designing a **modern, cross-platform management application**, Cameroon can eliminate bottlenecks, improve transparency, and enhance the experience for all stakeholders involved. This project represents a significant digital transformation opportunity for national pilgrimage management systems.

BOOK II : SPECIFICATION BOOK

Preambles

The Specification book is a document which is been established by the customer needing a particular product and the producer (engineer) who is to create the product, which is to be respected to its fullness during the realization of the product. From this, we obtain the major importance of the specification book.

INTRODUCTION

- I. CONTEXT AND JUSTIFICATION
- II. OBJECTIVES OF THE PROJECT
- III. EXPRESSION OF NEEDS
- IV. PROJECT PLANNING
- V. ESTIMATED COST OF THE PROJECT
- VI. PROJECT CONSTRAINTS
- VII. DELIVERABLES

CONCLUSION

INTRODUCTION

The specification book of our reports helps us provide details about our theme, to improve our understanding of it and increase the likelihood of it succeeding. To delimitate the scope of our project, we will specify the context of our theme. From the context, we will list the problems we have identified in our context and that we have decided to address throughout the project. After presenting our solution, we will talk about the objective we have set for the project. Also, we will explore the needs to which our system will respond both at the functional and non-functional level. We will then look at the estimated financial requirements for our project, and establish a plan we will follow to complete our project on time. From here we will discuss what is expected of us by the end of the project under the project deliverables

I CONTEXT AND JUSTIFICATION

CONTEXT

The Hajj pilgrimage is one of the most important spiritual obligations for Muslims around the world, including in Cameroon, where thousands of pilgrims participate annually. While the spiritual dimension of Hajj is immutable, the logistics behind organizing this mass movement of people require robust planning, modern tools, and institutional coordination. In Cameroon, Hajj operations are coordinated by the **National Hajj Commission**, working in collaboration with travel agencies, health services, and the Saudi Embassy. Currently, most administrative processes are conducted manually or via disparate systems, leading to:

- **Redundant paperwork**
- **Data inconsistencies**
- **Delays in processing**
- **Poor communication with pilgrims**
- **Lack of post-Hajj monitoring**

JUSTIFICATION

In response to all these challenges, the justification for this project lies in its ability to modernise the Hajj process through digitalization, leading to better efficiency and coordination. Given the increasing number of pilgrims and growing complexity of logistics, a **centralized digital platform** is both timely and necessary. The proposed system will help ensure **better coordination, data security, and real-time tracking** from registration to return. Moreover, this project aligns with Cameroon's national digital transformation goals, particularly within the public services and e-governance sectors. It also supports broader development objectives such as improving citizen engagement, promoting digital inclusion, and strengthening institutional performance.

II OBJECTIF OF THE PROJECT

Main Objective:

To design and develop a robust cross-platform management application for efficient coordination of Hajj operations in Cameroon.

Specific Objectives:

- Enable **online registration** of pilgrims with personal, medical, and travel details.
- Facilitate **biometric data collection** and **visa tracking**.
- Integrate **health screening** and **vaccination records**.
- Manage **travel itineraries**, flight schedules, and accommodation.
- Provide **real-time updates and notifications** to pilgrims.
- Track pilgrims' activities during Hajj and enable **emergency response support**.
- Conduct **post-Hajj surveys** and follow-up health assessments.
- Provide dashboards and analytics to administrators and stakeholders

III EXPRESSION OF NEEDS

Functional Needs

Functional needs describe what the users can do on the system. There are listed below as follows:

- Pilgrim registration portal (web and mobile)
- Admin dashboard for managing records
- Integration with health centers and visa systems
- Communication module (SMS/email/push notifications)
- Real-time tracking via mobile app
- Data export and analytics tools
- Offline support for field agents

Non-Functional Need

Non-functional requirements are a set of criteria that define the overall qualities, characteristics, and constraints of a software system, rather than its specific behaviors or functions. These requirements describe how the system should operate, rather than what it should do.

:Performance defines how fast a software system responds to certain user's action under certain workload. It explains how much a user must wait before the target operation

. **Reliability and Availability:**

The application should have high uptime and minimal downtime, ensuring users can access the platform consistently.

-The application should have a friendly user interface (UI) and should be easy to use

. -The code should be clear to facilitate future development and improvement

. **Security**

The application should provide a strong security mechanism to reassure user's that they can trust it for their sensitive information. Some of the security principles include confidentiality, Integrity and Authentication.

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Cross-platform compatibility

- **The application should be able to function on android, web and ios**
- Multilingual support it should be able to have these languages since they are the most spoken by the pilgrims (English, French,)

DELIVERABLES

In project management, any component materializing the result of a realization service is called a deliverable. In the case of our project, the deliverables are: A report composed of the following document :

- The application.
- The user guide.
- The PowerPoint.

IV PROJECT PLANNING

a. Chronogram of activities

Table 2 chronogram of activities

PHASE	OBJECTIVE	OUTPUT	DURATION PERIOD	
INSERTION	Collecting informations from concerned entities	Insertion book	3 days	1 th to 3 th July 2026
EXISTING SYSTEM	Study of the existing system	Existing System	3 days	6 to 8th July 2026
SPECIFICATION BOOK	Specification of the user needs	Specification Book	5 days	9 to 15 July 2026
ANALYSIS	ANALYSIS	Capture of needs Use case and textual description Modelling	1 weeks	16 to 24 July 2026
CONCEPTION	Hardware and software conception	Conception Book	2weeks	27 July to 13 August 2026
REALIZATION	Realization of the system	Realization Document	3weeks	14 August to 11 September 2026
DEPLOYMENT	Deployment of the system	Testing and debugging the software	1week	14 to 22 th September 2026
WRITING USER GUIDE	Instruction on how to use the software and hardware	User Guide	2 days	23 th to 24 th September 2026

GANTT Project

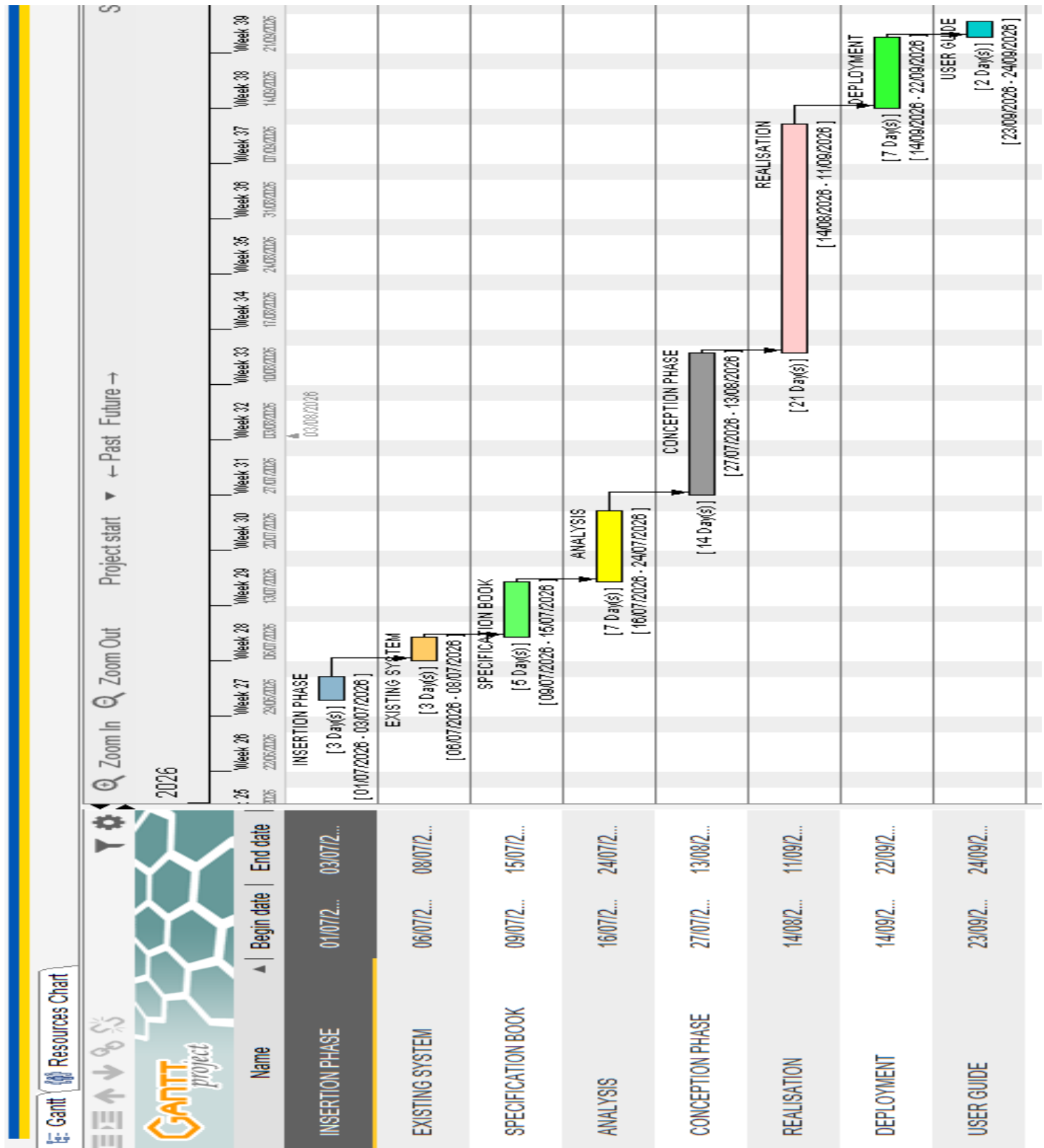


Figure 3:gantt chart

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V ESTIMATED COST OF THE PROJECT

a. Software's Resources

Table 3 software estimation Mercurial 2026

RESOURCES	DESIGNATION	USAGE	QUANTITY	UNIT COST(FCFA)
Formatting	Microsoft office 2016	Writing Report	1	287,500
Cloud storage	GitHub	Saving report and source code	1	Freemium
Web browser	Google chrome	For research and viewing the application	1	Freeware
Illustrating tool	Edrawmax	For drawing the geographical location of the Enterprise	1	Freemium
UML analysis	Visual Paradigm	For drawing UML diagrams	1	Freemium
Project planning	Gantt Project	For building Gantt chart	1	Freemium
Code Editor	Visual studio code	For writing code of the applications	1	Freemium
Database	MySQL	To store different informations	1	Freemium
TOTAL 1				287,500

b. Hardware Resource -source

Table 4hardware resources (mercurial 2026)

Resources	Usage	Quantity	Unit cost (FCFA)
HP ProBook 450G2	For drafting the report and coding	1	150,000
Printer HP	For printing the report	1	447 350
Modem (TOP-LINK)	For network connectivity	1	50,000
USB key 64GB	For installation of and sharing of software	1	30, 490
TOTAL2			677,840

c. Human Resources

resources(<https://www.paylab.com/cm/salaryinfo/information-technology.>)

Table 5human resources

ROLES	NUMBERS DAYS	OF QUANTITY	COST PER DAY (IN FCFA)	TOTAL PRICE (IN FCFA)
Project Manager	90	1	40,000	3,600,000
Analyst	30	1	30,000	900,000
Designer	07	1	20,000	140,000
Programmer	30	2	15,000	900,000
Tester	14	1	10,000	140,000
Unforeseen				1,000,000
Total 3				6,680,000

d. Global Estimation

Table 6Global estimation

TOTAL1 (IN FCFA)	TOTAL2 (IN FCFA)	TOTAL3 (IN FCFA)	OVEROLL TOTAL (IN FCFA)
287,500	879,890	6,680,000	7,847,390
SEVEN MILLION FIFTY THOUSSAND SEVEN HUNDRED AND FIFTY			

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VI PROJECT CONSTRAINTS

Project constraints are the limiting factors that affect the project's success. They often fall into several categories like

Budget Constraints

- **Limited funding:** Especially if funded by a religious organization, NGO, or government body with restricted financial resources.
- **Cost of development tools:** While open-source tools may reduce costs, integration, hosting, security, and maintenance still have expenses.
- **Training and support:** Post-development training for users (pilgrims, agents, administrators) adds to the cost.

Time Constraints

- **Pilgrimage season deadline:** The system must be ready before the Hajj season begins each year, creating a fixed development window.
- **Time for testing:** Adequate time must be allocated for alpha and beta testing before launch.
- **Government or religious deadlines:** These may impose strict timelines for development milestones.

Scope Constraints

- **Feature limitations:** Due to budget or time, not all desired features (e.g., real-time GPS tracking, multi-language support, or biometric verification) may be included initially.
- **Integration boundaries:** Integrating with external systems like immigration, airlines, or health databases may not be fully possible.
- **User roles:** Scope must clarify how many roles (pilgrim, travel agent, admin, health worker, etc.) are supported in version 1.

. Human Resource Constraints

- **Skilled personnel:** Shortage of local developers with experience in cross-platform mobile development or cloud systems.
- **Stakeholder availability:** Gathering requirements from multiple stakeholders (pilgrims, Hajj organizers, government bodies) may be delayed.
- **User training needs:** Many users (especially older pilgrims) may be digitally illiterate.

VII LIST OF PARTICIPANTS AND DELIVERABLES

A LIST OF PARTICIPANTS

Table 7list of participants

NAME	FUNCTION	ROLE
Mr. YVANN EMATE	Supervisor at SHADERL	Professional supervisor at SHADERL
Mr. NDENGE	Lecturer at AICS-Cameroon	Academic supervisor
HAWA ADAMA MAMOUDOU	Intern at SHADERL	Student at AICS-Cameroon

CONCLUSION

The successful implementation of this integrated Hajj management system will greatly improve operational efficiency, reduce fraud, enhance pilgrim satisfaction, and allow better coordination between government agencies, service providers, and pilgrims. It represents a major step toward **digital transformation** in religious travel management in Cameroon, with potential for long-term scalability and regional adaptation

BOOK III. ANALYSIS BOOK

Preamble

The main objective of the analysis phase is to capture the user's need, the delimitation of the field of study and to have a clear understanding of the system in study. To achieve this, we will use UML (Unified Modelling Language) with the 2TUP (2 Track Unified Process) as method applied to UML to analyse the system. We will start by doing a comparison between UML and MERISE, of various unified processes and lastly, we will present the modelling of the solution we propose (Our software).

Content

INTRODUCTION

I. METHODOLOGY

1. COMPARATIVE STUDY OF UML AND MERISE.

2. COMPARATIVE STUDY OF UNIFIED PROCESSES.

II. CHOICE OF THE ANALYSIS METHOD

3. MODELLING OF THE PROPOSED SOLUTION CONCLUSION

INTRODUCTION

System development can be thought of as having two major components: System analysis and system design which both help in understanding the details of the existing system or the system to be designed. The analysis and design of information systems has most of the time vocation to allow the creation of databases, which must represent as closely as possible the reality of the field studied thus requiring the use of a design method. This is why our choice will be directed on the UML method as it offers much to developers seeking a user-centered approach and / or a wide scope in design. This part of the report consists of the comparative study of UML and MERISE, unified processes and finally the various diagrams that meet the functional need requirements.

I. METHODOLOGY

1. COMPARATIVE STUDY OF UML AND MERISE

a. MERISE

MERISE stands for “Méthode d’Etude et de Réalisation Informatique pour des Systèmes d’Entreprise”. Although it is prescriptive to some extent, MERISE permits the participation of end users and senior management as well as data processing professionals in its decision cycle. MERISE is a method for designing, developing and carrying out IT projects. The goal of this method is to achieve the design of an information system. The MERISE method is based on the separation of data and processing to be carried out in several conceptual and physical models. The essentials of the approach lie in its three cycles: the decision cycle, the life cycle and the abstraction cycle, which cover data and process elements equally. The separation of data and processing ensures longevity in model. Indeed, the arrangement of data does not have to be often overhauled, while treatments are more frequently.

b. UML

UML (Unified Modelling Language) is a standard notation for the modelling of real world objects as a first step in developing an object-oriented design methodology. Its notation is derived from three object-oriented design and analysis methodologies: Grady Booch's methodology for describing a set of objects and their relationships, James Rumbaugh's Object-Modelling Technique (OMT), Ivar Jacobson's approach which includes a use case methodology. Other ideas also contributed to UML, which was the result of a work effort by Booch, Rumbaugh, Jacobson, and others to combine their ideas, working under the sponsorship of Rational Software. UML captures information about the static and dynamic view of a system. UML 2.5 comprises of 14 diagrams which represent the different views of a system. The 14 diagrams can be subdivided into two, Static or structural and Dynamic diagrams. These diagrams include

• STATIC OR STRUCTURAL DIAGRAMS

- ❖ Class diagram;
- ❖ Object diagram;
- ❖ Component diagram;
- ❖ Deployment diagram;
- ❖ Composite Structure diagram;
- ❖ Package diagram;
- ❖ Profile Diagram;

. BEHAVIOURAL OR DYNAMIC DIAGRAMS

- ❖ Use case diagrams;
- ❖ Activity diagram;
- ❖ State machine diagram;
- ❖ Sequence diagram;
- ❖ Communication diagram;
- ❖ Interaction diagram;
- ❖ Timing Diagram;

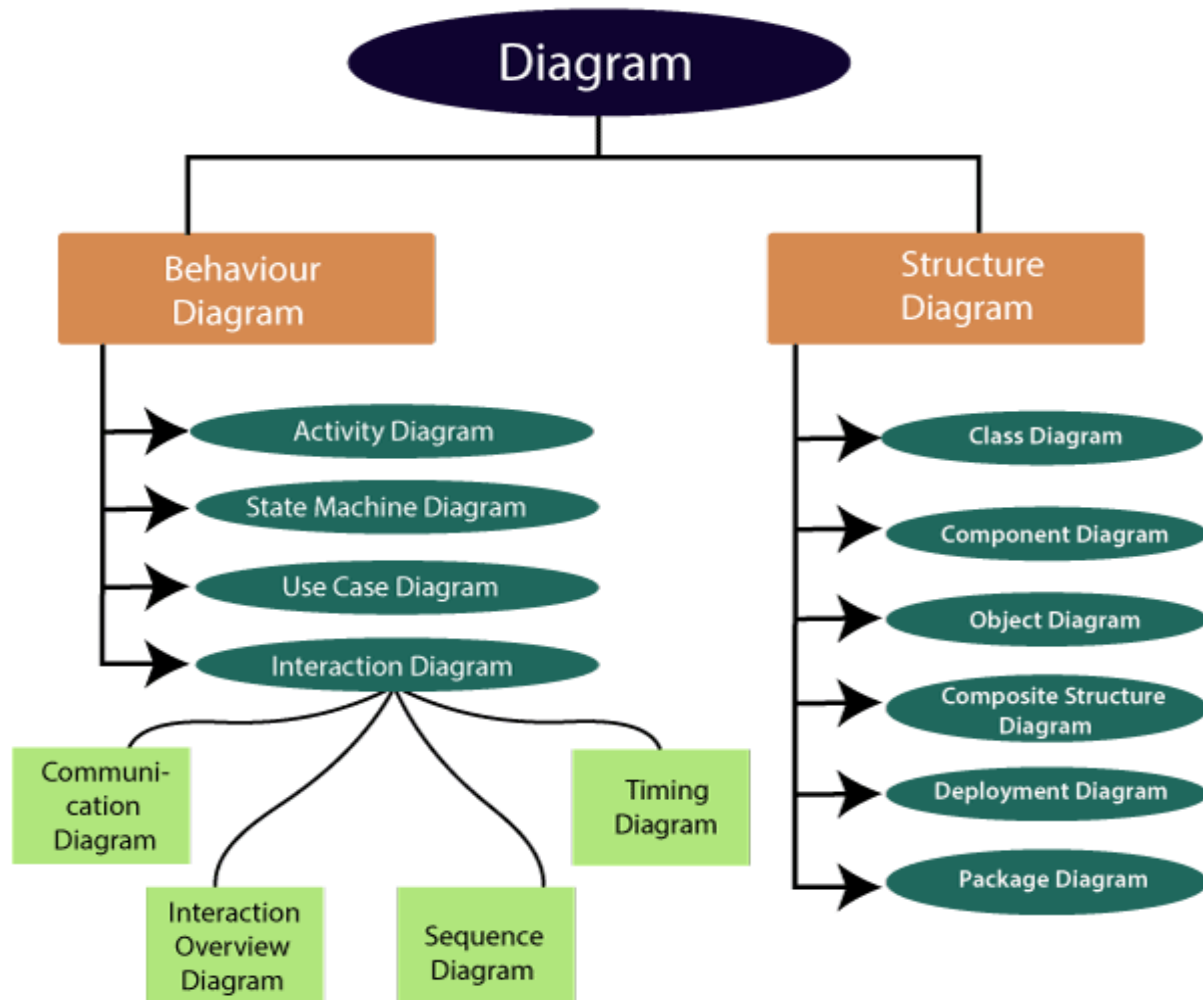


Figure 4Structure of UML

Table 8 difference between UML and MERISE

MERISE	UML
It stands for Méthode d'Étude et de Réalisation Informatique pour les Systèmes d'Entreprises	Unified Modelling Language
MERISE is a systemic method of analysis and design of information systems. That is, it uses a systems approach.	UML is however not a method but rather an object modelling language to which it is necessary to associate an approach to make it a method. This is the case with the 2TUP method; RUP and XP.
MERISE proposes to consider the real system from two points of view: - A static view (data) - A dynamic view (treatments). That is, with the MERISE method, we have a separate study of the data and the treatments.	UML offers a different approach from that of MERISE in that it combines data and processing. Because with UML, centralizing the data of a type and the associated processing makes it possible to limit the maintenance points in the code and facilitates access to information in the event of software development. In addition, UML describes the dynamics of the information system as a set of operations attached to the objects of the system.
Rational	Object

2. COMPARATIVE STUDY OF UNIFIED PROCESSES

a. A Unified Process

A Unified Process is a process of development of software constructed on UML; it is iterative, incremental, centered on architecture, driven by use cases and requirements.

Iteration is distinct sequence of activities with a basic plan and evaluation criterion that produces an internal or external output. Either the content of an iteration is improved, or the evolution of the system is evaluated by users.

An increment is the difference between two released products at the end of two iterations. Each iteration that the group is capable of integrating the technical environment in order to develop a final product and give users the possibility of having tangible results.

Centered on architecture the different models derived during the establishment of system must be reliable and coherent.

Driven by use case and requirements enables the clear definition of a users' needs and priorities respectively thereby minimizing the risk of project failure.

b. The Two Track Unified Process (2 TUP)

2TUP is a unified process which is built on UML and has as objective to bring solution to constraints of functional and technical changes imposed on information systems by strengthening controls on development capacities. It proposes a Y-shaped development life cycle that separates the functional aspect from the technical aspects, and the merging of these two forms the implementation aspect. 2TUP distinguishes therefore two branches: the functional and technical branches, the combination of the result of these two branches forms the third: the realization branch – where we realize our system. The diagram below illustrates the branches of 2TUP.

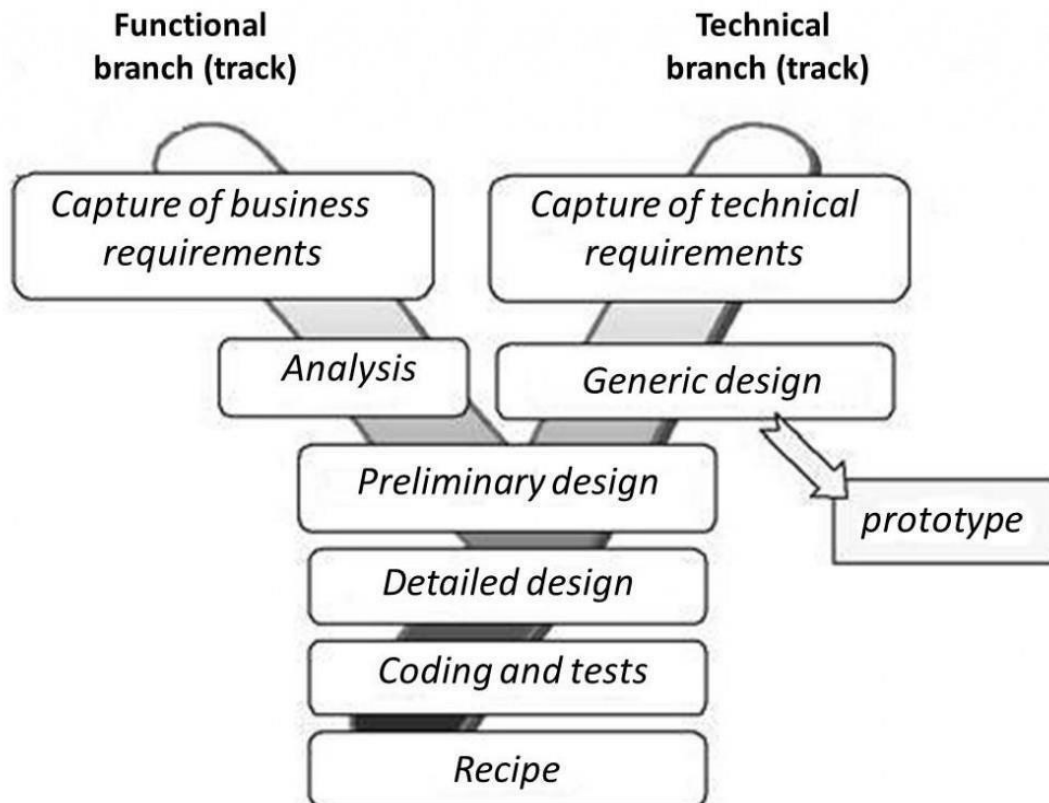


Figure 52TUP diagram

The left branch (functional branch)

It captures the functional needs of a system. This ensures the production of software that meets the needs/requirements of the user. The analysis here consists of studying precisely the functional specification in order to obtain an idea of what the system is going to realize, and its result does not depend on any technology.

The right branch (Technical branch)

The technical branch enumerates the technical needs and proposes a generic design validated by a prototype. The technical needs include constraints and choices related to the conception of the system, the tools

The middle branch (Realization or Implementation branch)

In this branch, we study the preliminary conception, detailed conception, and documentation of the system. The realization branch supports the following:

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Preliminary conception: This is the most sensitive step of 2TUP as it is the confluence of the functional and technical branch. It is completed when the deployment model, the operating model, the logical model, interphases and the software configuration model are defined. We have the following diagrams:

- + Component Diagram;
- + Deployment Diagram;
- + Package Diagram;
- + Composite Structure Diagram;

Detailed conception: This is the detailed design of each feature of the system. We have the following diagrams:

- + Class;
- + Sequence;
- + Object;
- + Timing diagram;

Coding and testing: This is the phase where we program the designed features and test the coded features.

The recipe: Also known as the deliverables is the validation phase of the functions of the developed system.

I. MODELLING OF THE PROPOSED SOLUTION

A. Capture of Functional Needs

The first step of the left (functional) branch of Two Track Unified Process (2TUP) is the capture of the functional needs. At this step, we capture the intended behavior of the system that maybe express as services, tasks or functions the system is required to perform.

1. USE CASE DIAGRAM

a) Definition

Use-case diagrams describe the high-level functions and scope of a system. These diagrams also identify the interactions between the system and its actors. The main objectives of the use case diagram are:

- Provide a high-level view of the system.
- Identify the functions of the system.

b) Formalism

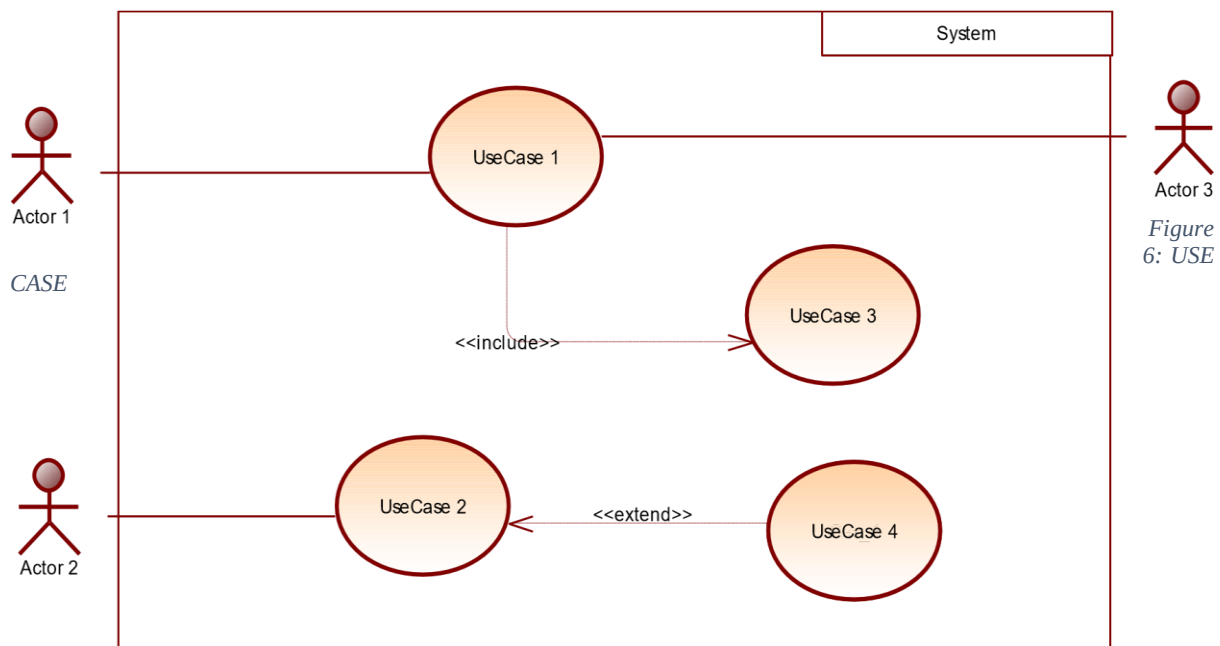
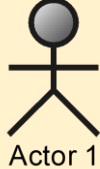
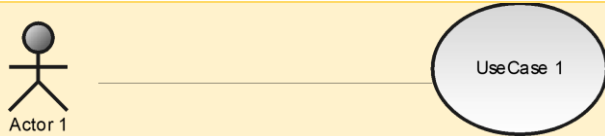
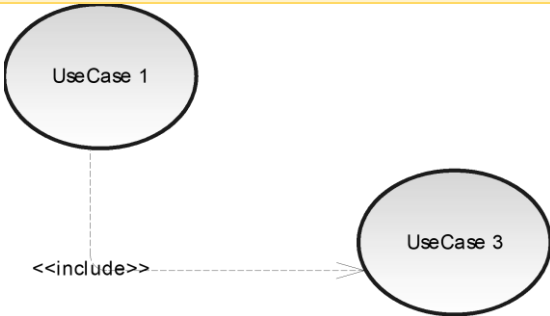
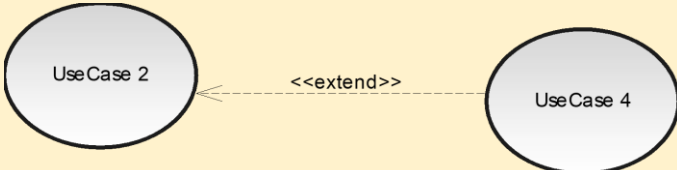


Figure 6: USE

c) Components of a use case diagram

Elements	Description	Description
Actors		Represents an entity that directly interacts with the system. The actor is what performs the different possible actions of the system
Use case		A use case represents a functionality of the system. It is an action that can be performed by an actor
Association		it indicates that an actor takes part in a use Case.
Include		An inclusion denotes that an included action must be performed before the including action can be performed
Extend		An extension denotes that an extending action may be performed while an extended action is being performed.

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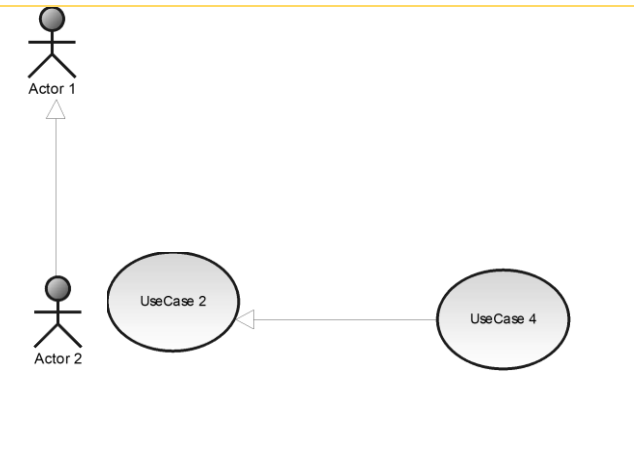
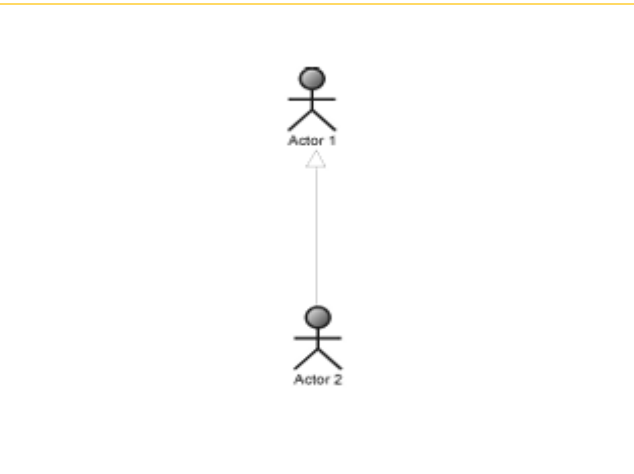
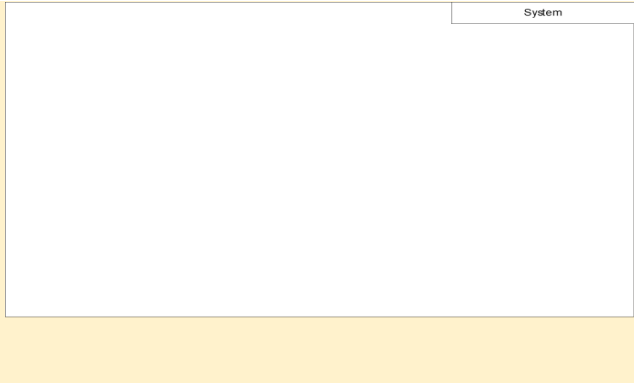
Generalization		This shows that an actor or a use case is a kind of another abstract or concrete actors can be defined and later specialized using generalization relationship
Inheritance		It is the only possible relation between actors
System		It is a container of use cases which interact with external actors

Table 9: Components of a use case diagram

System Actors

Actors	Role
Administrator	Manages accounts Manage courses Manage system
Pilgrim guide	manage Notifications Manage courses manage profile
Pilgrim	Manage profile Make payment
Verification API	Verifies pilgrim's judiciary status
Payment API	Receive payment for the orientation session

Graphical representation of use cases diagrams

General use case

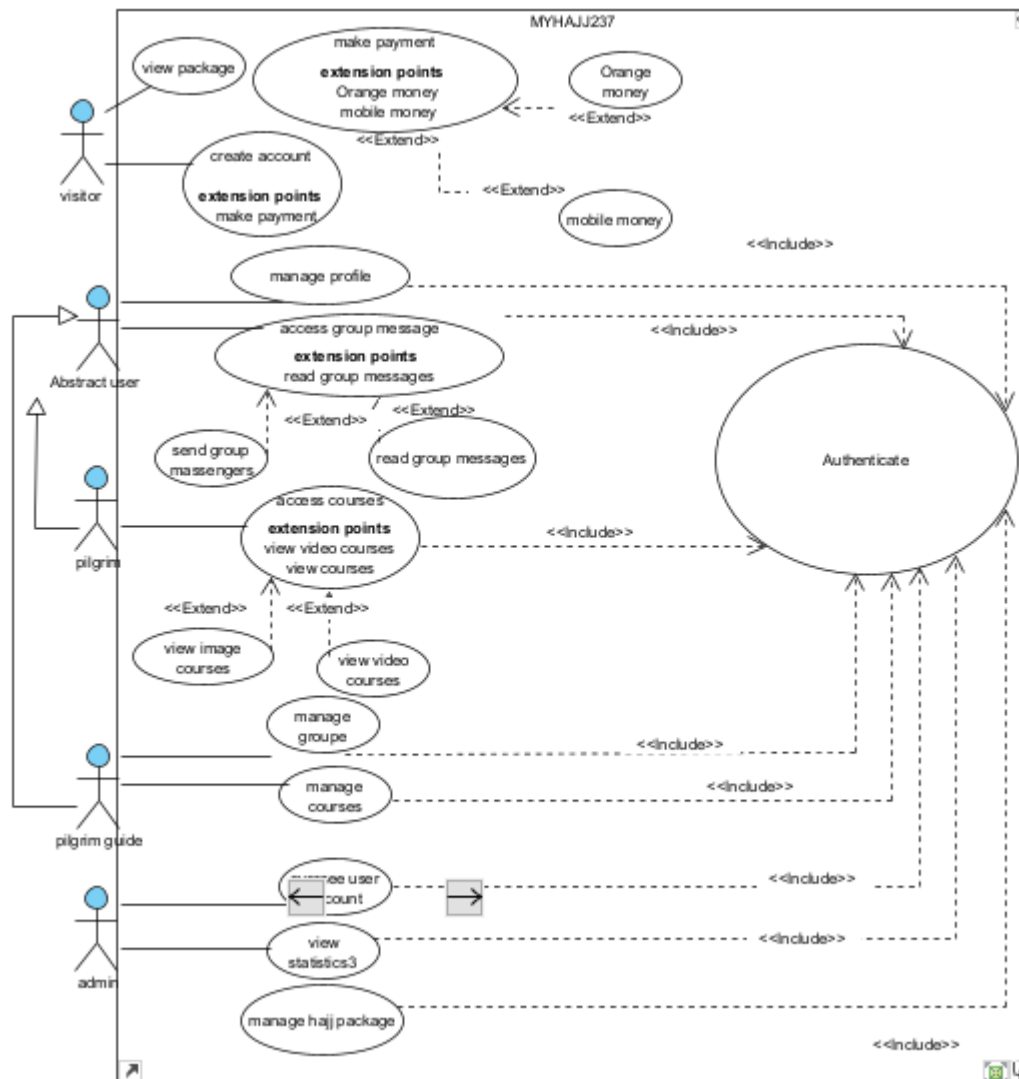


Figure 7: General USE CASE diagram

Specific use case manage group

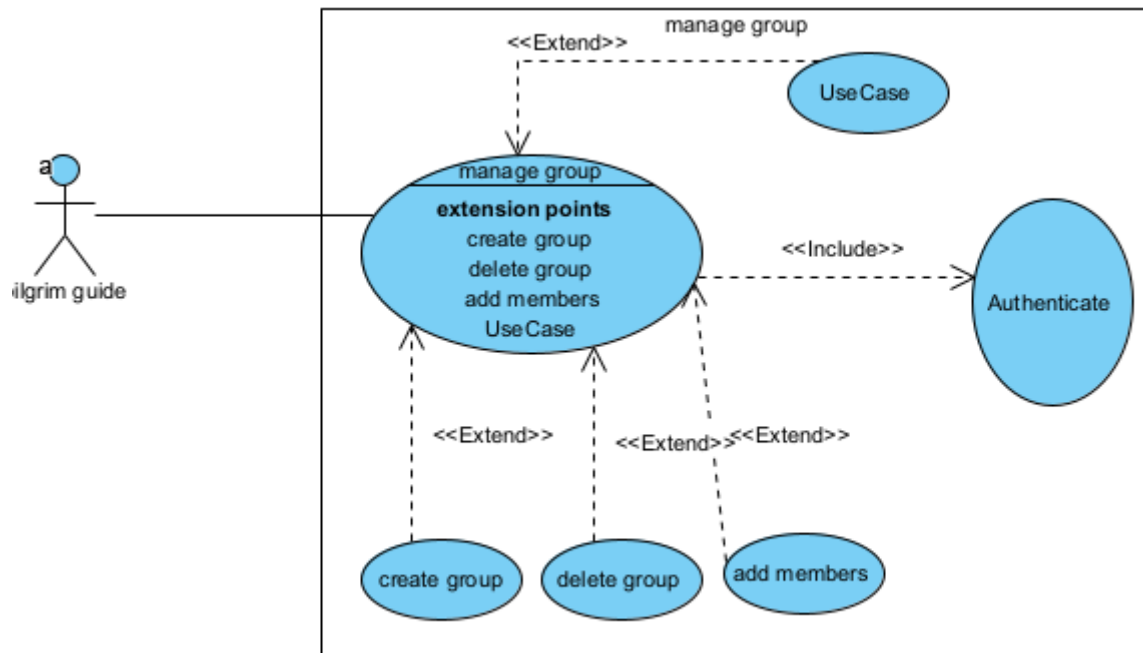


Figure 8: USE CASE diagram "Manage courses"

f. Specific use case manage courses

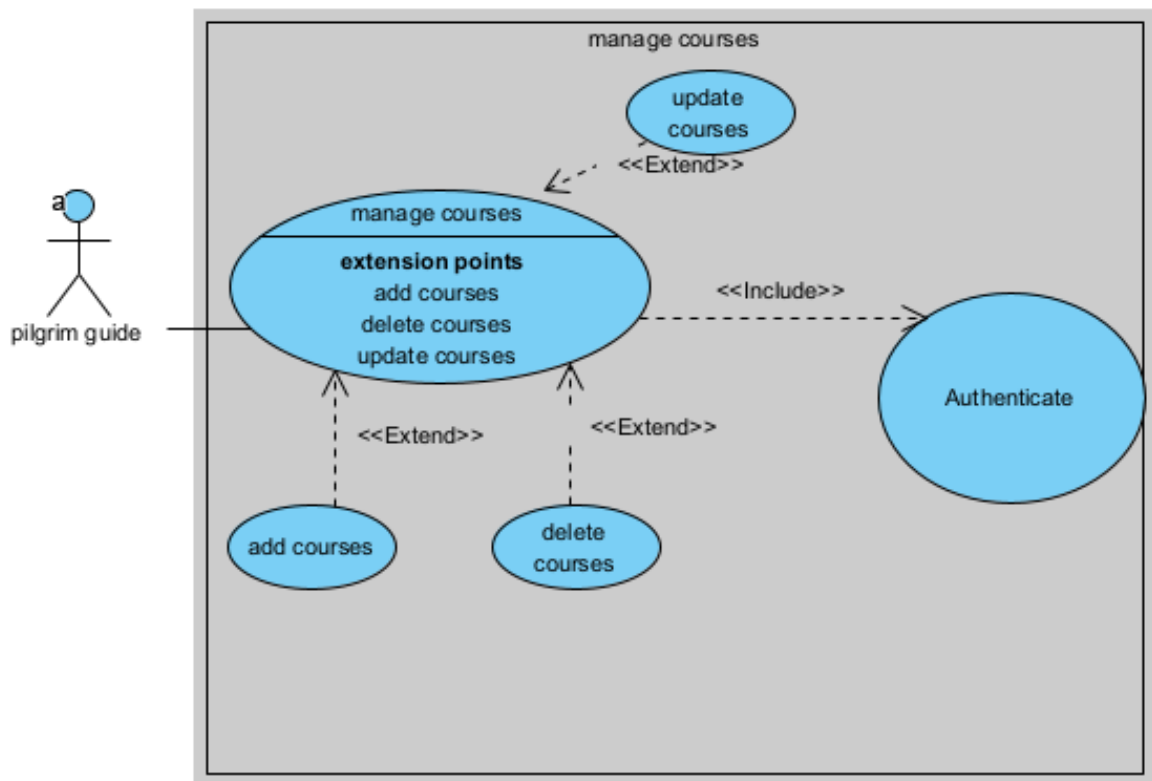


Figure 9: USE CASE diagram "Manage courses"

USE CASE textual description

As we mentioned earlier, a use case represents a transaction. Here, it is a question of breaking down the use cases, making a textual description and then emphasizing on the different detailed scenario. This will be done using diagrams known as the sequence and activity diagrams.

a- Use case “ Create an account”

Table 10: Textual description Register

Title	Authenticate
Summary	The user needs to authenticate
Actors	Pilgrim, pilgrim guide
Date	1 st September
Version	2.5
Precondition (s)	1. The app is launched. 2. The actor has an account on the platform.
Triggers	The user clicks on login button
Nominal Scenario	1. The system displays the login form. 2. The actor fills and submits the form. 3. The system verifies conformity of the form fields 4. The system sends the data to the dbms. 5. The dbms returns result of the query. 6. The system displays a success message to the actor.
Alternative Scenarios	7. At step 4 of the nominal scenario, the user enters mismatched or missing information. 8. The system displays an error message then returns to step 2 of the nominal scenario.
Post condition of success	The user has access to his/her dashboard
Post condition of failure	The user does not have access to the platform
Non-functional requirement	Entering the password must not be visible on the screen

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Delete course textual description

Table 11:delete course textual description

Title	Delete course
Summary	The user needs to delete a course
Actors	pilgrim guide
Date	1 st September
Version	1.0
Precondition (s)	The pilgrim guide authenticate
Triggers	The user clicks on delete button
Nominal Scenario	1.The system displays the available courses. 2.The pilgrim guide select a course and click delete buttton. 3.The system displays running message 4.The pilgrim guide click on delete . 5.The system displays a success message to the actor.
Alternative Scenarios	6.At step 1, if the system is unable to display course page 7.The system displays an error message 8.At step 5, if the system is unable to delete a course, 9.The system display error message
Post condition of success	The system displays success message
Post condition of failure	The message is not deleted

Create group textual description

Table 12:create group textual description

Title	Delete course
Summary	The user needs to create a group
Actors	pilgrim guide
Date	1 st September
Version	1.0
Precondition (s)	The pilgrim guide is authenticated
Triggers	clicks on create group button
Nominal Scenario	1.The system displays groups. 2.The pilgrim guide click on create group button. 3.The system displays create group form 4.The pilgrim guide fill and click on create group 5.The system displays a success message to the actor.
Alternative Scenarios	6.At step 1, if the system is unable to display group page 7.The system displays an error message 8.At step 5, if the pilgrim guide doesn't fill the form correctly, 9.The system display error message and returns form
Post condition of success	The system displays new group created
Post condition of failure	The pilgrim guide, is unable to create a group

Communication diagram

i- Definition

In UML, a communication diagram shows the interactions between the objects or roles associated with lifelines and the messages that pass between lifelines. They describe both the static structure and the dynamic behavior of a system. It is a simplified version of a Collaboration Diagram introduced in UML 2.0.

ii- Formalism

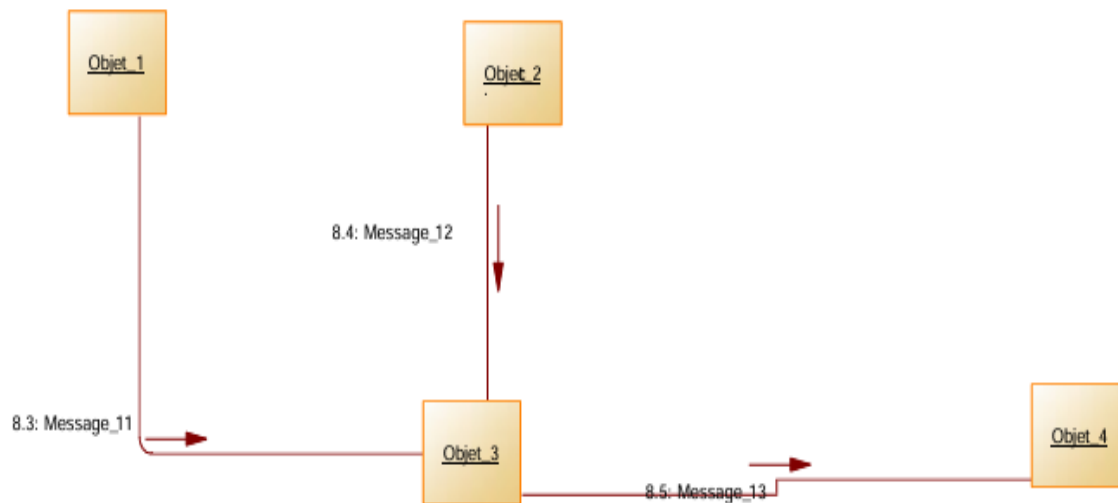
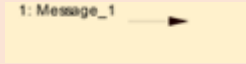
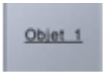


Figure 10: Communication diagram formalism

a) Components of communication diagram

Table 13: Components of communication diagram

ELEMENT	NOTATION	DESCRIPTION
Message		Designs a particular communication between lifelines
Connectors		It represents the relationship that exist between lifeline
Dependency		A dependency is a relationship that signifies a single or a set of model elements for their specification
lifeline		An object represents an individual participant in the interaction

b) Authentication communication diagram

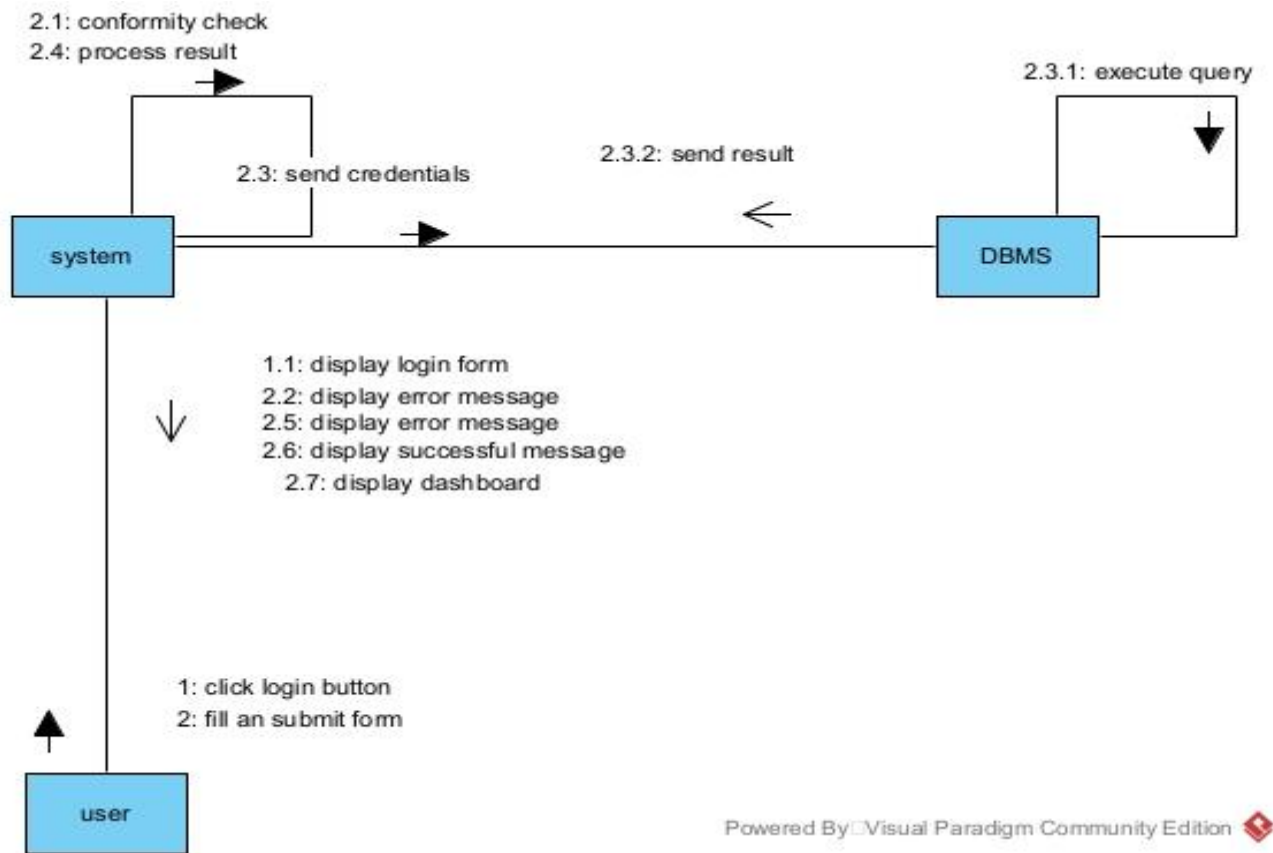


Figure 11: Authentification communication diagram

c) Create group communication diagram

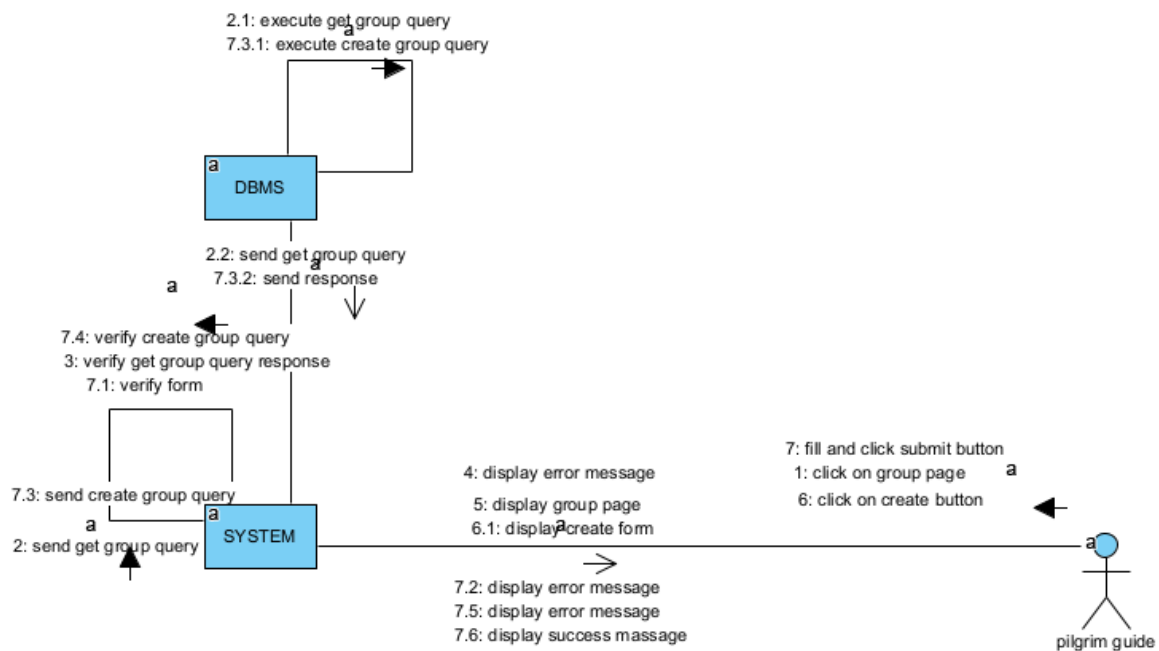


Figure 12: create group communication

d) Delete course communication diagram

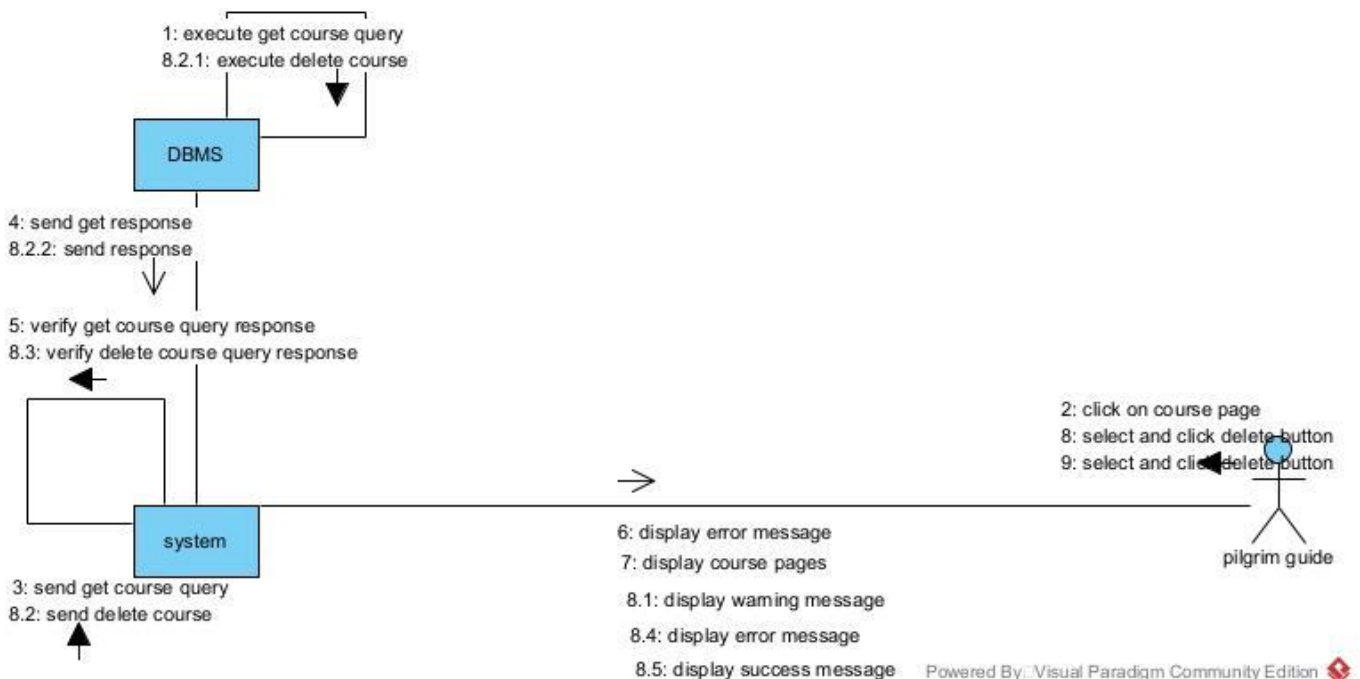


Figure 13: delete course communication

SEQUENCE DIAGRAM

a) Definition

A sequence diagram in UML (Unified Modelling Language) is a type of interaction diagram that shows the sequence of events or interactions between objects or components in a system. It illustrates how objects communicate with each other,. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modelling a new system.

b) Formalism

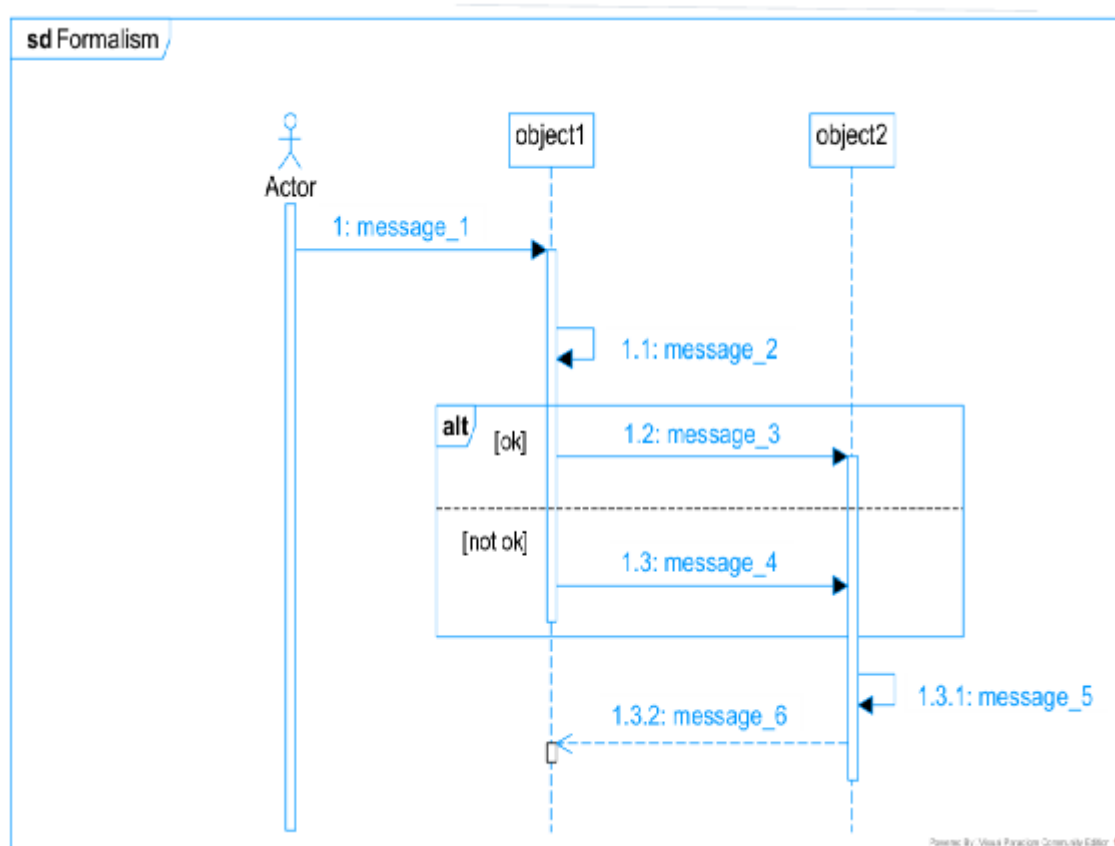
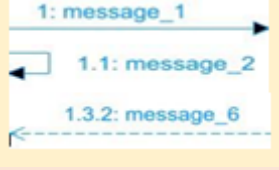


Figure 14: Sequence diagram formalism

Table 14: Components of sequence diagram

ELEMENTS	NOTATION	DESCRIPTION
Lifeline		An individual participant in a sequence diagram, is positioned at the top of the diagram
Combined fragment		It represent a choice of behaviour in which at most one operand will be chosen
Messages		These are arrows which shows the direction of message flow we have the synchronous, the self-message and the asynchronous
Activation		It describes the time period in which an operation is performed by an element

c) Authentication sequence diagram

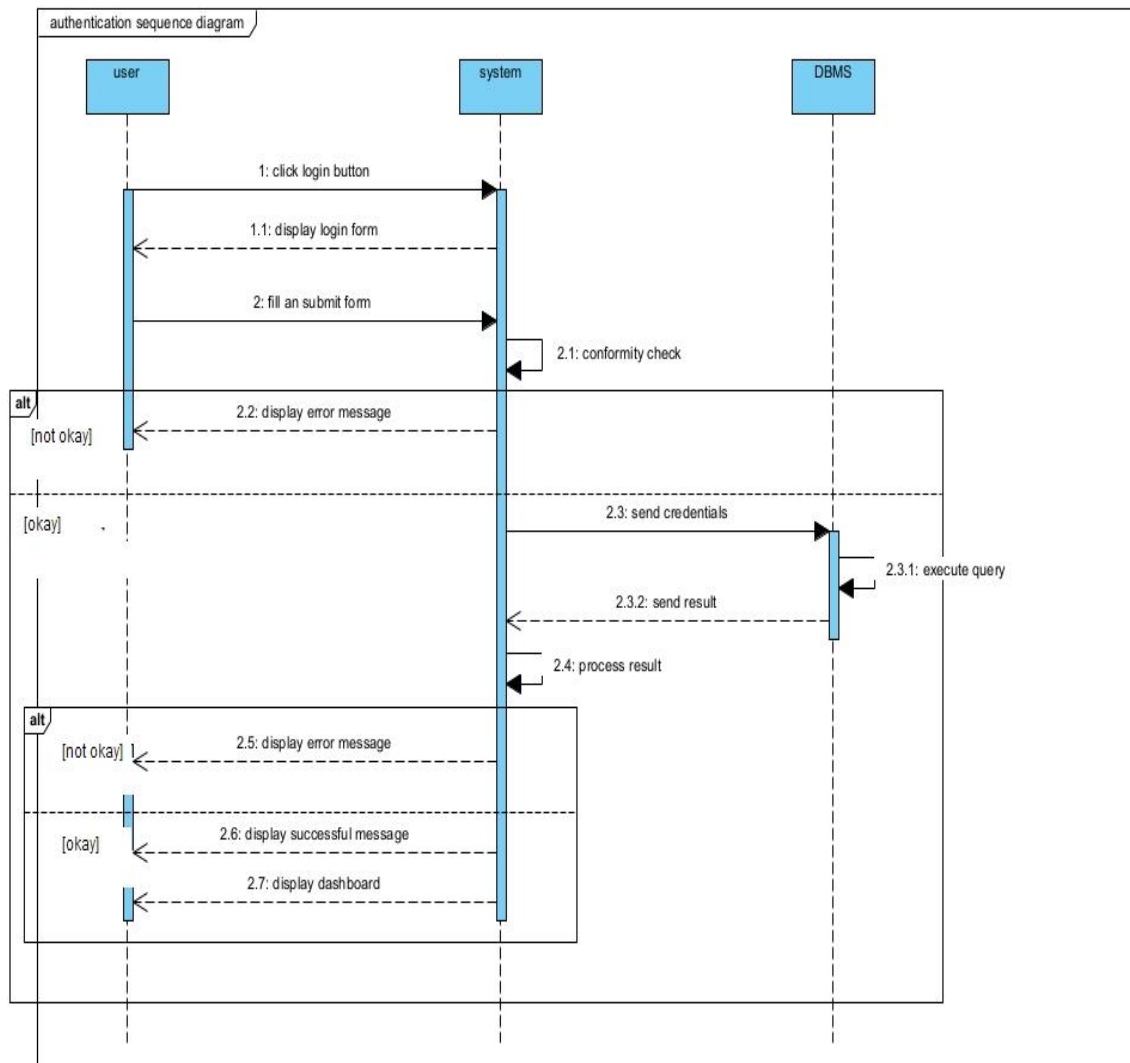


Figure 15: Authentication sequence diagram

d) Delete course sequence diagram

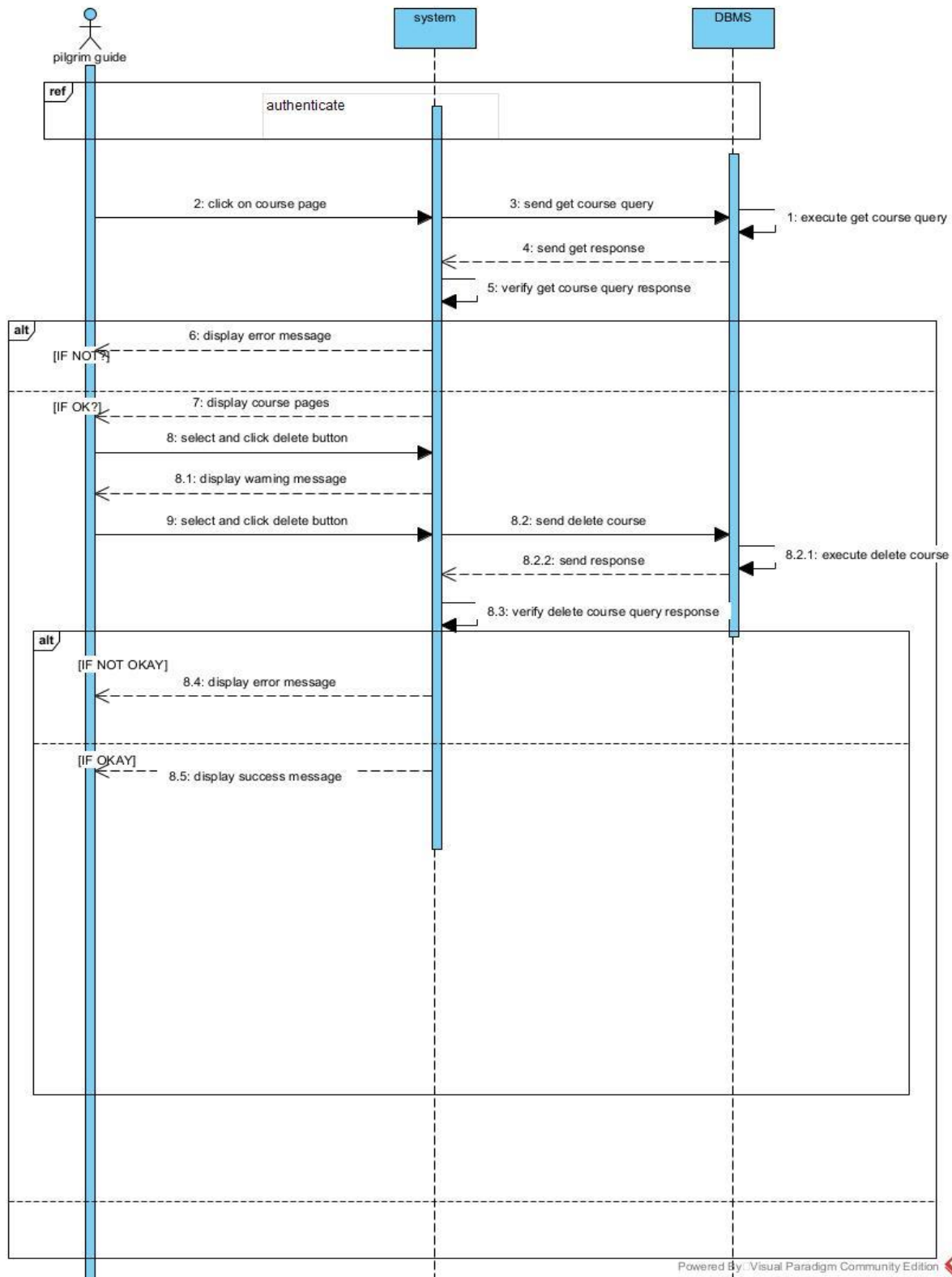


Figure 16:delete sequence diagram

d) Create group sequence diagram

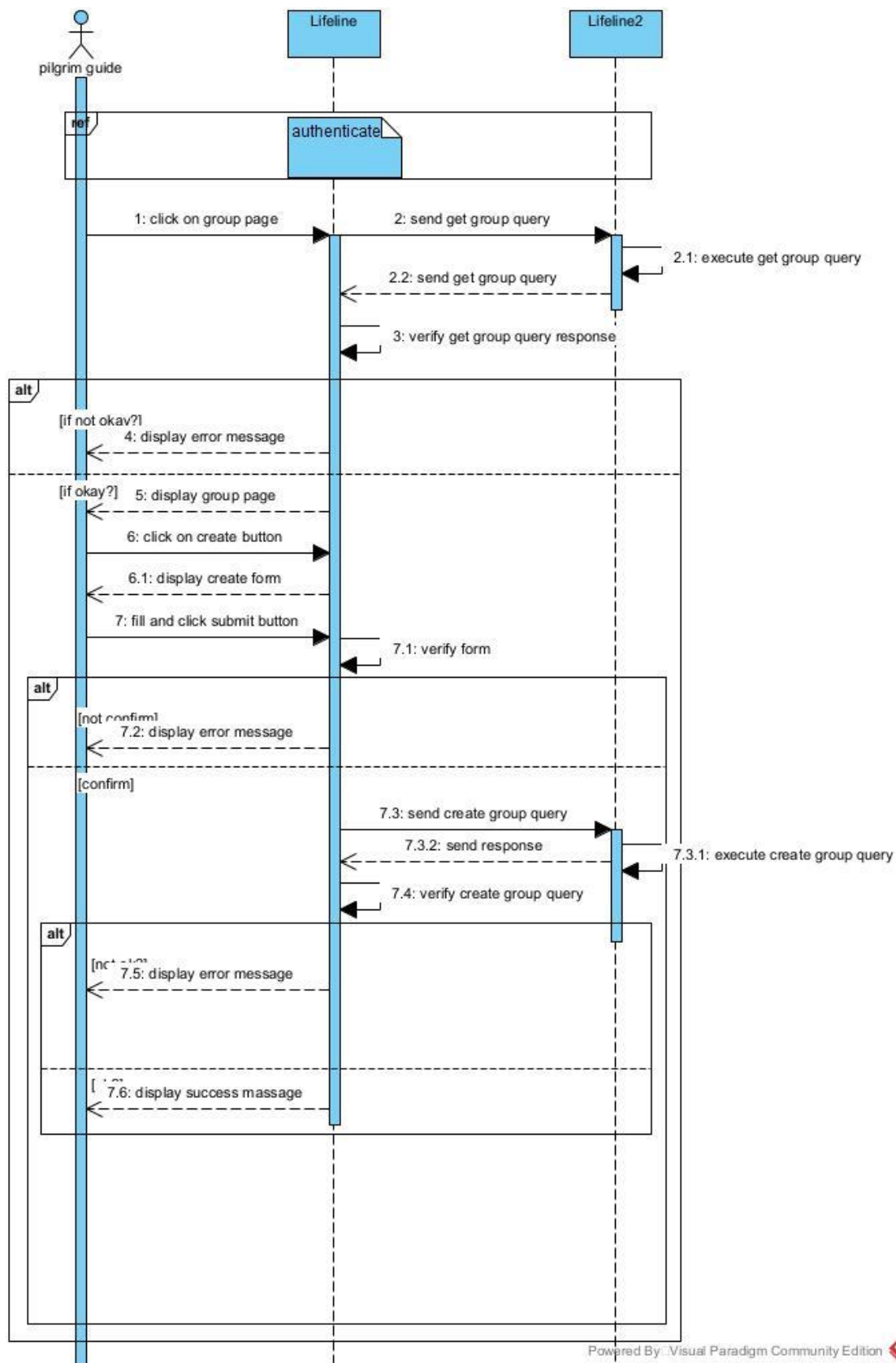


Figure 17: Create group sequence diagram

Written by **HAWA ADAMA MAMOUDOU,**

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BIYA TECHNOLOGICAL CENTER OF EXCELLENCE Academic year 2025-2026

ACTIVITY DIAGRAM

a) Definition

An activity diagram is a graphical representation of workflows that show the steps needed in the realization of a process; showing the details from a start point to an end point through all decisions and actions that can possibly be performed. Activity diagrams are intended to model both the computational and organizational process. They flow can be sequential, branched or concurrent. Below is an activity diagram formalism.

b) Formalism

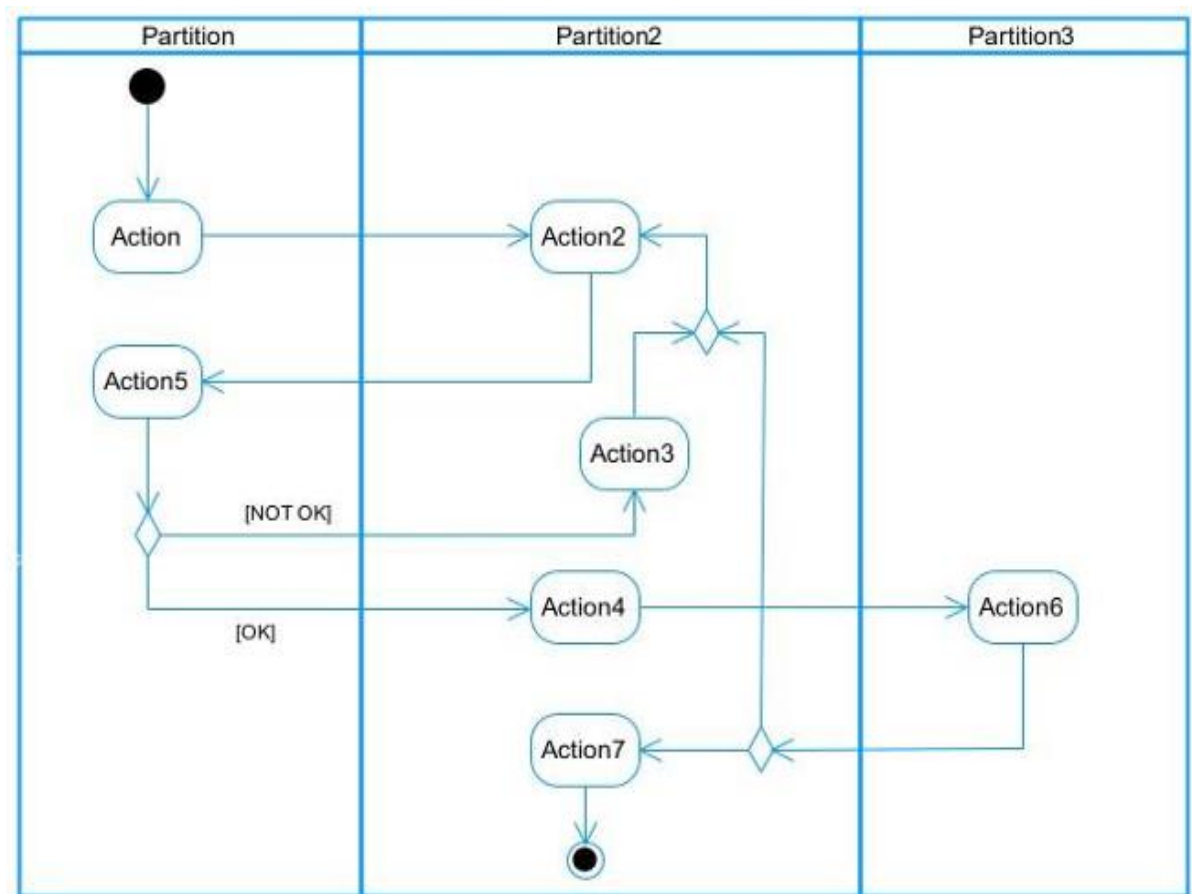
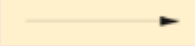
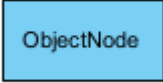


Figure 18:formalism of activity diagram

Table 15: Components of activity diagram

Element	Diagrammatic representation	Description
Activity		Used to represent a set of actions
Action		Represents a task to be performed
Activity edge		A directed connection between two activity nodes through which tokens may flow
Initial node		Shows the beginning of an activity or set of actions
Final node		Stops all control and object flows in an activity
Object node		Represents an object connected to a series of object flows
Decision node		Represents a test condition that splits an incoming activity edge into opposite outgoing activity edges

a) Register activity diagram

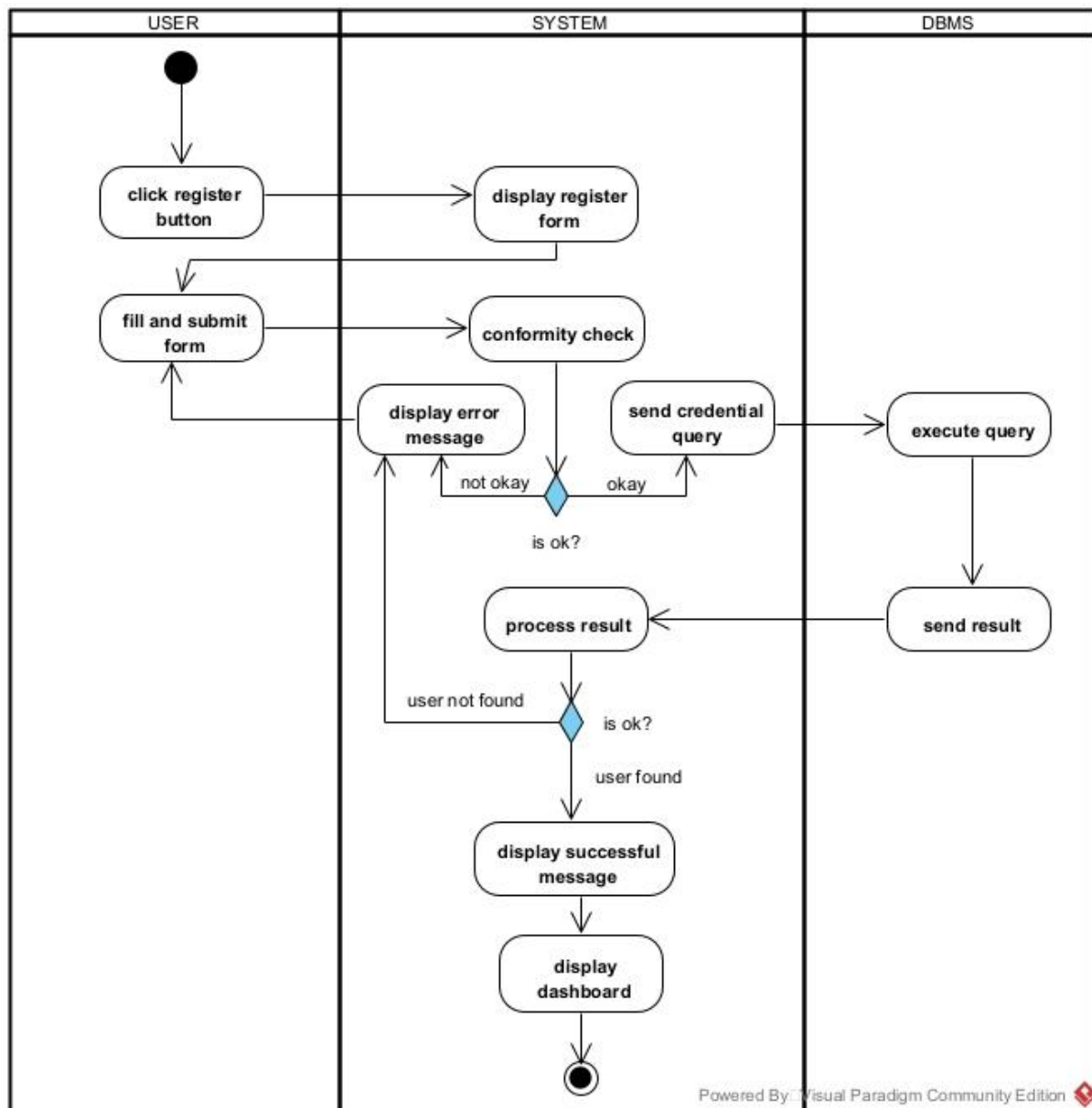


Figure 19:register activity diagram

b) Create group activity diagram

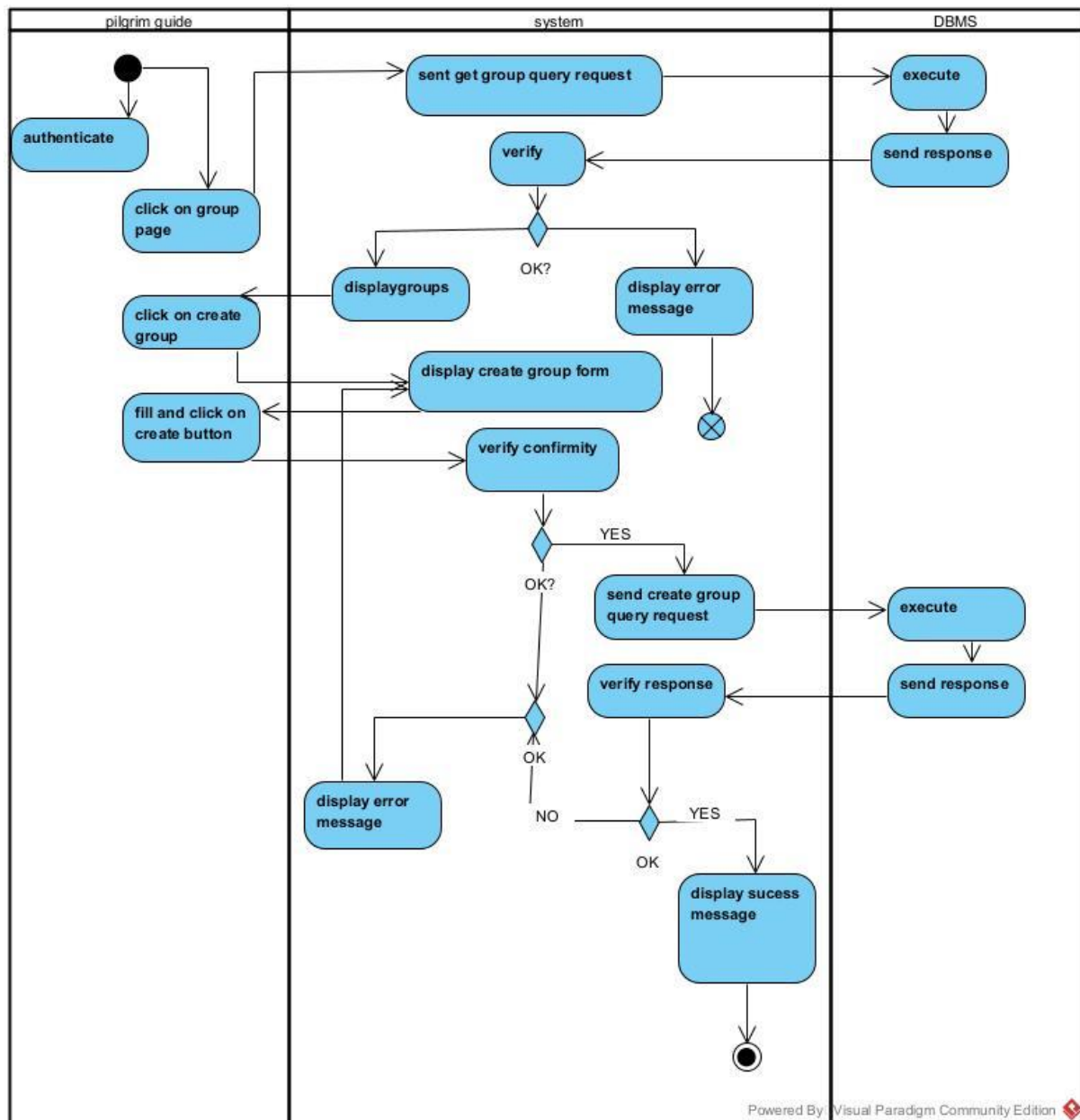


Figure 20: create group activity

c)Delete course activity diagram

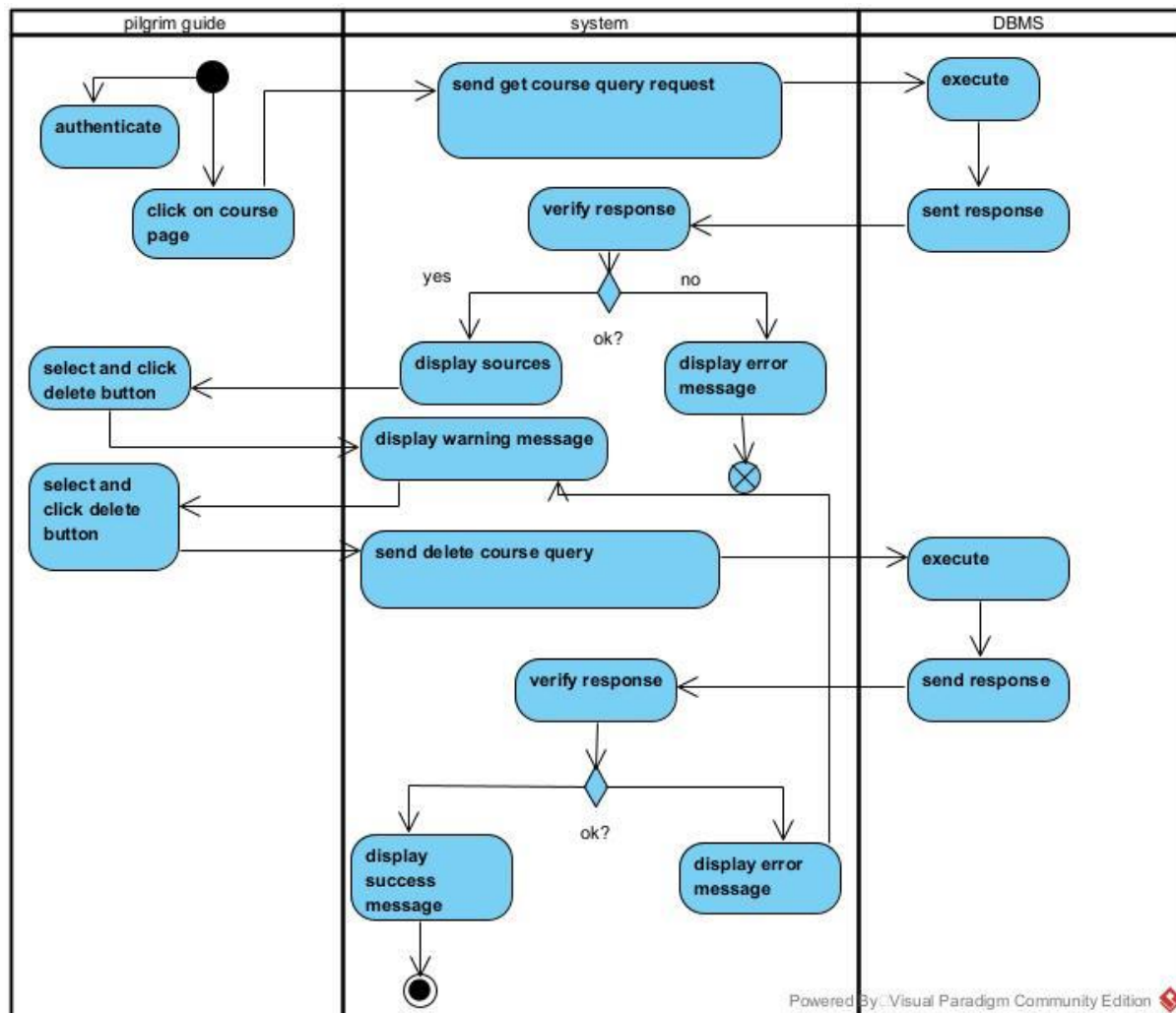


Figure 21:delete course activity

CONCLUSION

In the analysis phase, we chose a software development process and modelling language, after which we explained the functional need of our system, we saw the use case diagram which shows the relationship between the actors and use case (the action the actor can perform on the system), we saw the communication diagram which represents the architecture of the system based on object oriented programming, we saw the sequence diagram which represents the flow of messages between elements in the system, and lastly the activity diagram which shows the workflow of our system. We will now move to the conception phase in which we will present the Technical branch of our system together with related diagrams.

BOOK IV : CONCEPTION PHASE

Preamble

The conception phase will permit us to present in an orderly manner the components necessary for the good functioning of our software and also the architecture used for the proposed solution. It bridges the gap between the analysis phase and the realization phase.

Content

INTRODUCTION

- I- CLASS DIAGRAM
- II- STATE MACHINE DIAGRAM
- III- PACKAGE DIAGRAM

CONCLUSION

INTRODUCTION

The conceptual phase will describe in details the necessary specifications, features and operations that will satisfy the functioning requirements of the proposed system as modelled in the analysis phase. This phase is meant to identify and consider essential components (hardware /or software), structure (network capabilities), processes and procedures for the system to accomplish its objectives. We will look at some diagrams such as the component diagram, package diagram, communication diagram, class diagram and the object diagram.

CLASS DIAGRAM

Definition

A class diagram is a static diagram. It represents the static view of an application. class diagram is not only used for visualizing, describing and documenting different aspect of the system but also for constructing executable code of the software application. Class diagram describes the attribute and operation of a class and constraints imposed on the system. Its purpose is to model the static view of an application.

b. Formalism

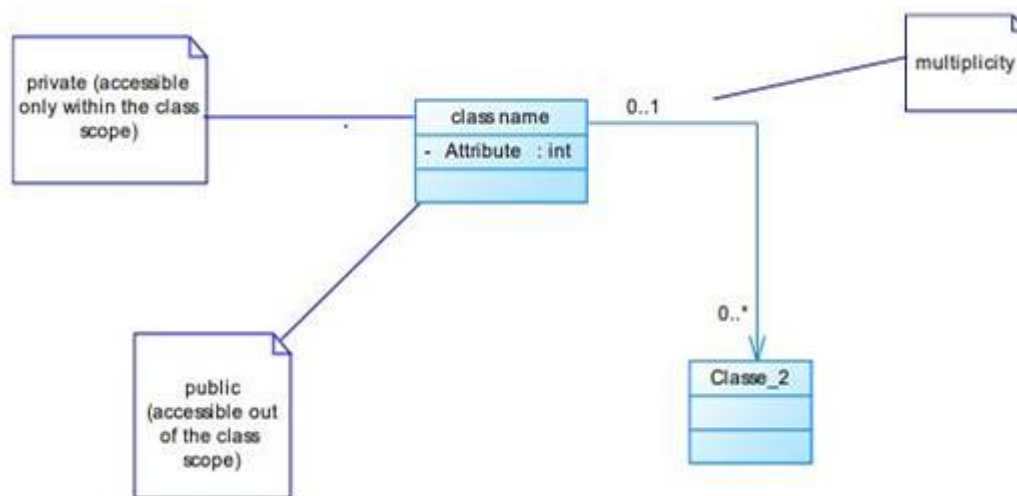
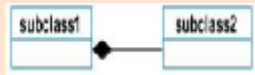
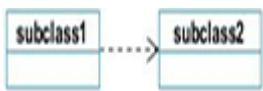
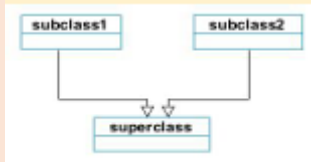


Figure 22:formalism of a class diagram

Components of class diagram

Element	Representation	Decription
Class		A class is an element that defines that attributes and behaviours that an object is able to see generate
Composition		If a parent of a composite is deleted usually all of its parts are deleted with it
Aggregation		If the parents of the aggregate is deleted usually the children are not deleted
Dependency		It existed between two classes, i one changes it may cause the change in order, but the other way around
Generalisation		It is a type of relationship between two classes where one class (child class) inherits the properties of the other class (parents class)
Association		It is a general type of relationship between elements it may include cardinality, roles etc

system class diagram

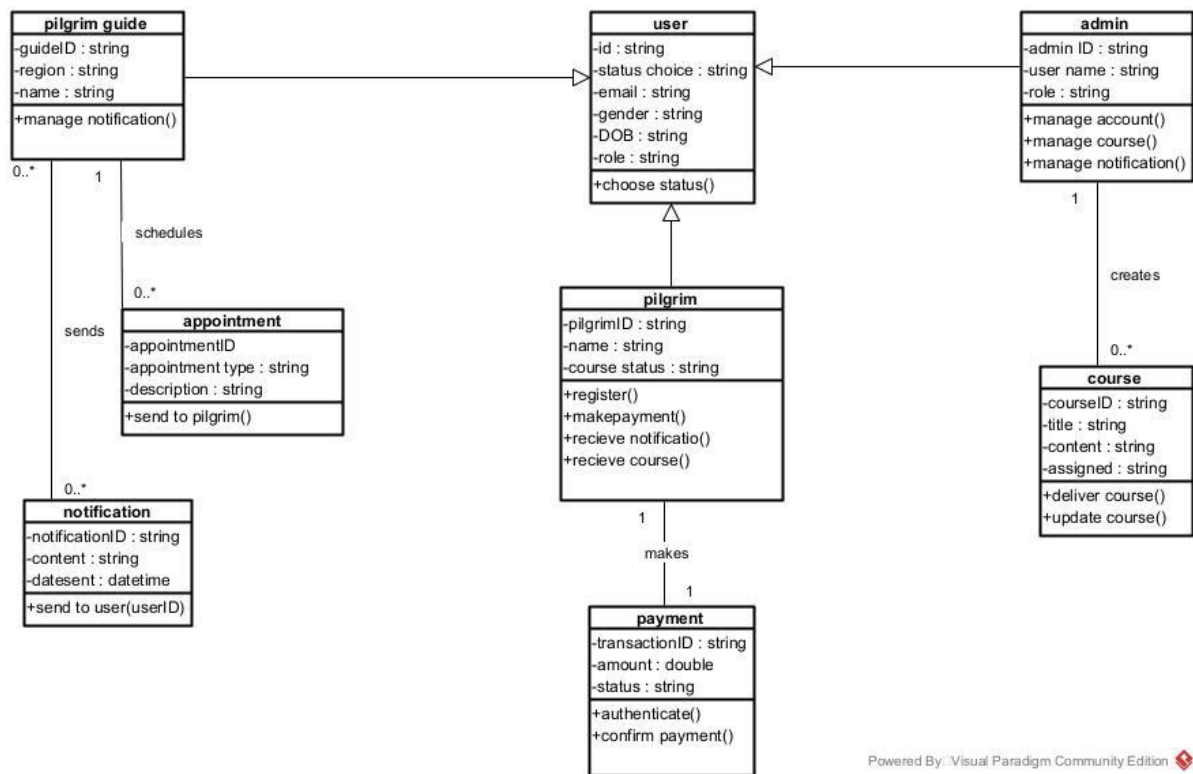


Figure 23: System class diagram

STATE MACHINE DIAGRAM

Definition

A state machine diagram describes the behavior of a single object in response to a series of events in a system. Also known as the state machine diagram, it models the dynamic flow of control from the state of a particular object within a system.

a. Formalism

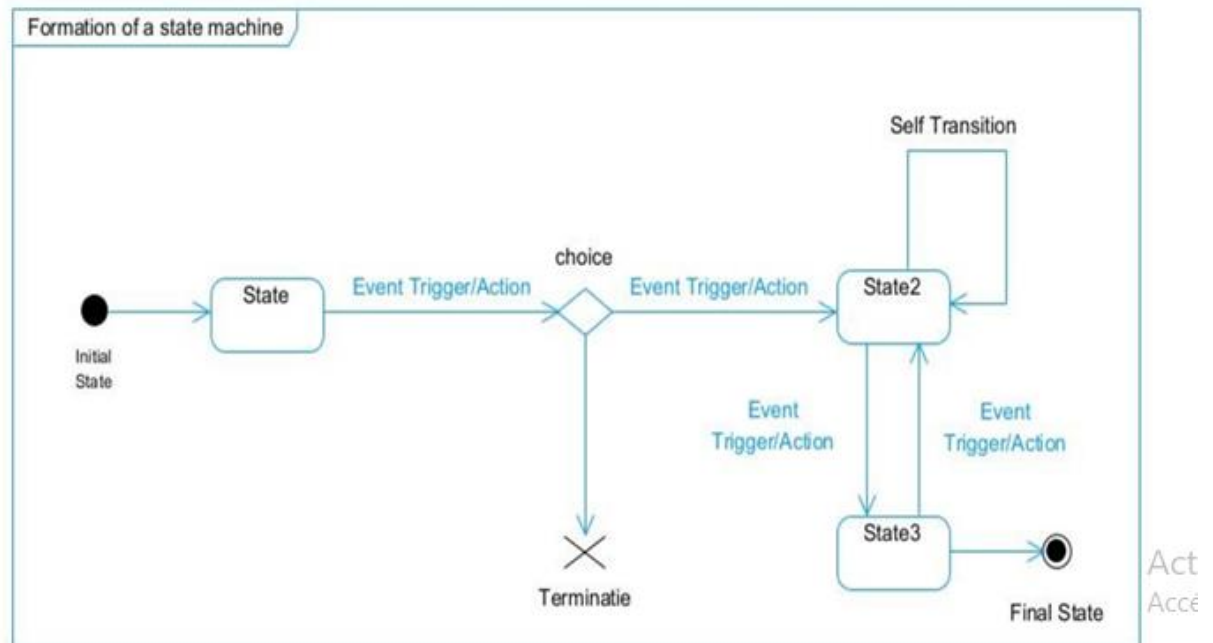
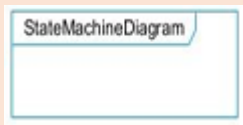


Figure 24:formalism of statemachine

Table 16: Components of a statemachine

Components of a statemachine

element	Notation	Description
State		Models situation during which a certain invariant condition holds
First (initial) state		It represents a default vertex, that is a source for a single transaction to the default or composite state
Final state		A state specifying that the enclosing region is complete
Transition		A direction relation between a source and a target vertex
Choice pseudo state		A diamond symbol that indicates a dynamic condition with branches potential results
Terminate		Implies that the execution of a state by means of it context is terminated
Diagram Overview		A placeholder for the linked states in a state machine diagram

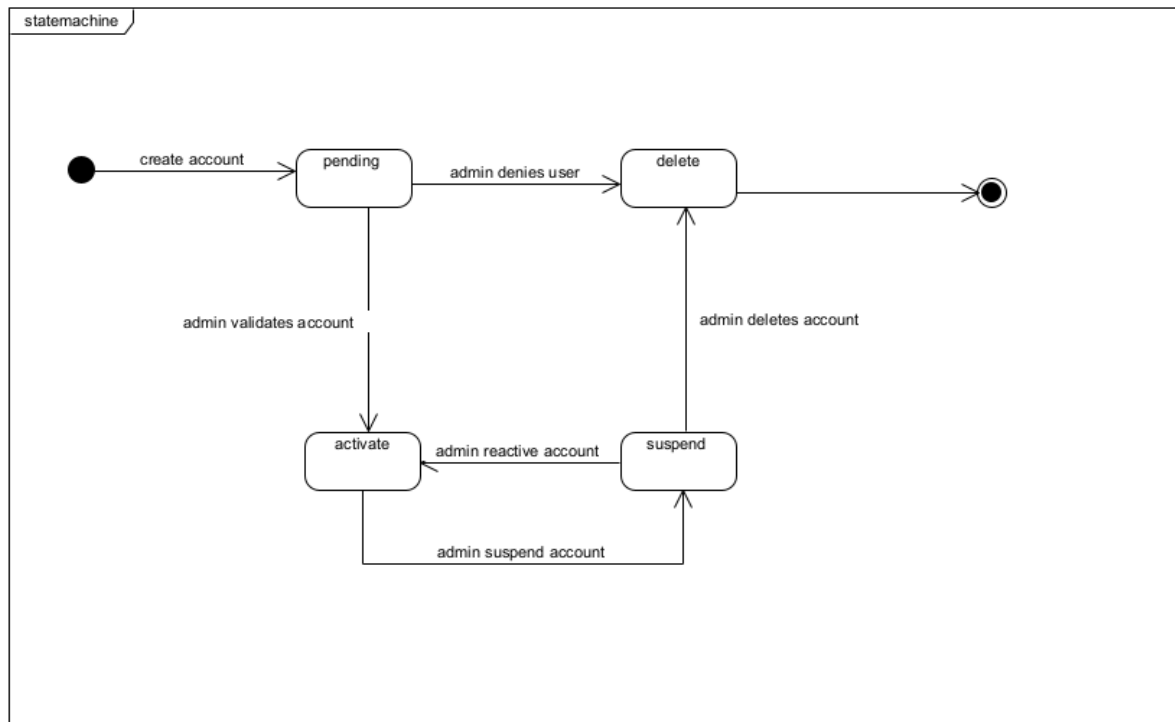


Figure 25:system statemachine

PACKAGE DIAGRAM

Definition

This is a structural diagram used to show the organization and arrangement of various model elements in the form of packages. A package diagram is the grouping of related uml elements such as classes, diagrams or even other packages.

b. Formalism

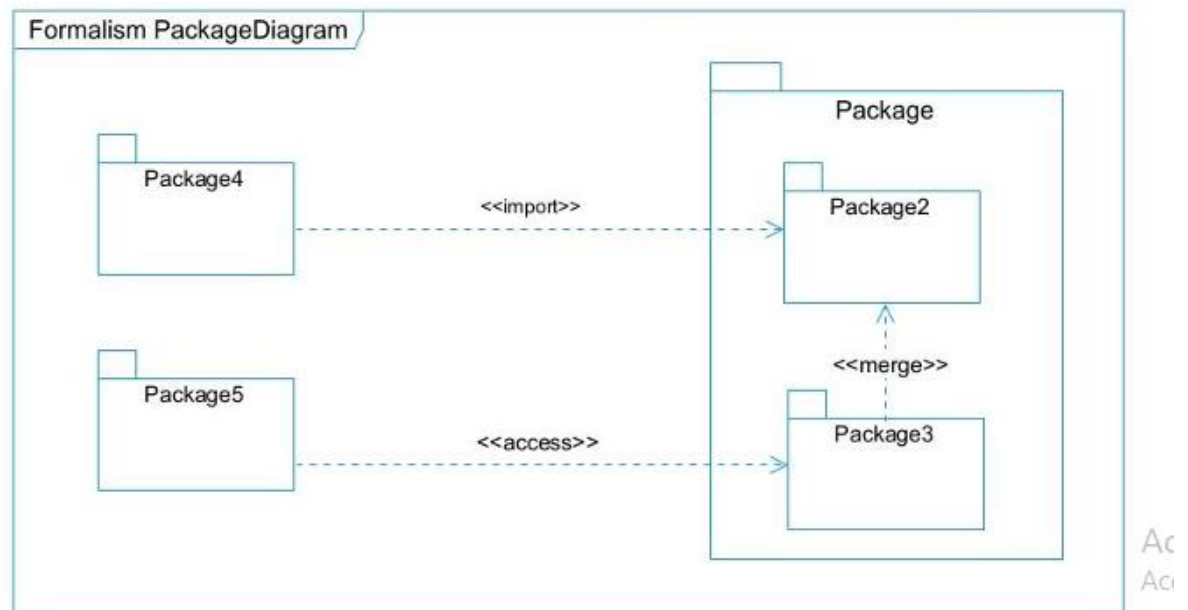
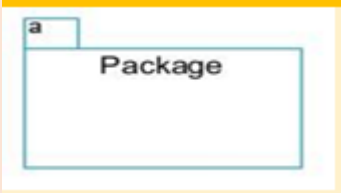
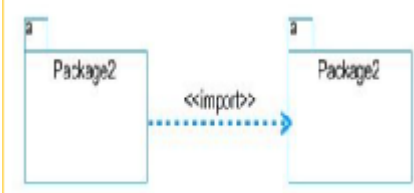
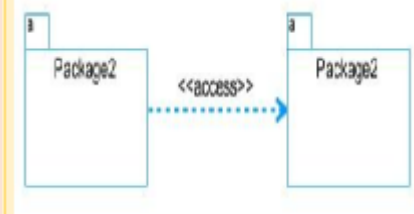
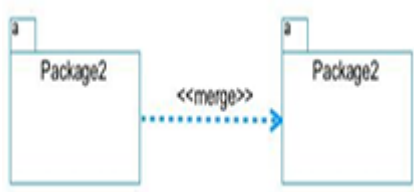


Figure 26:formalism of package

Component of package

Table 17 component of package diagram

Name	Representation	Description
Package		A package is a namespace use to group related elements; it is a mechanism used to group elements into a better structure in a system
Package import		A relationship indicate that functionality has been imported from one package to another
Package access		A relationship indicates that one package requires assistance from the function of one another package
Package merge		It is a relationship which shows that the functionality of two packages are combines to a single function

Package diagram of system

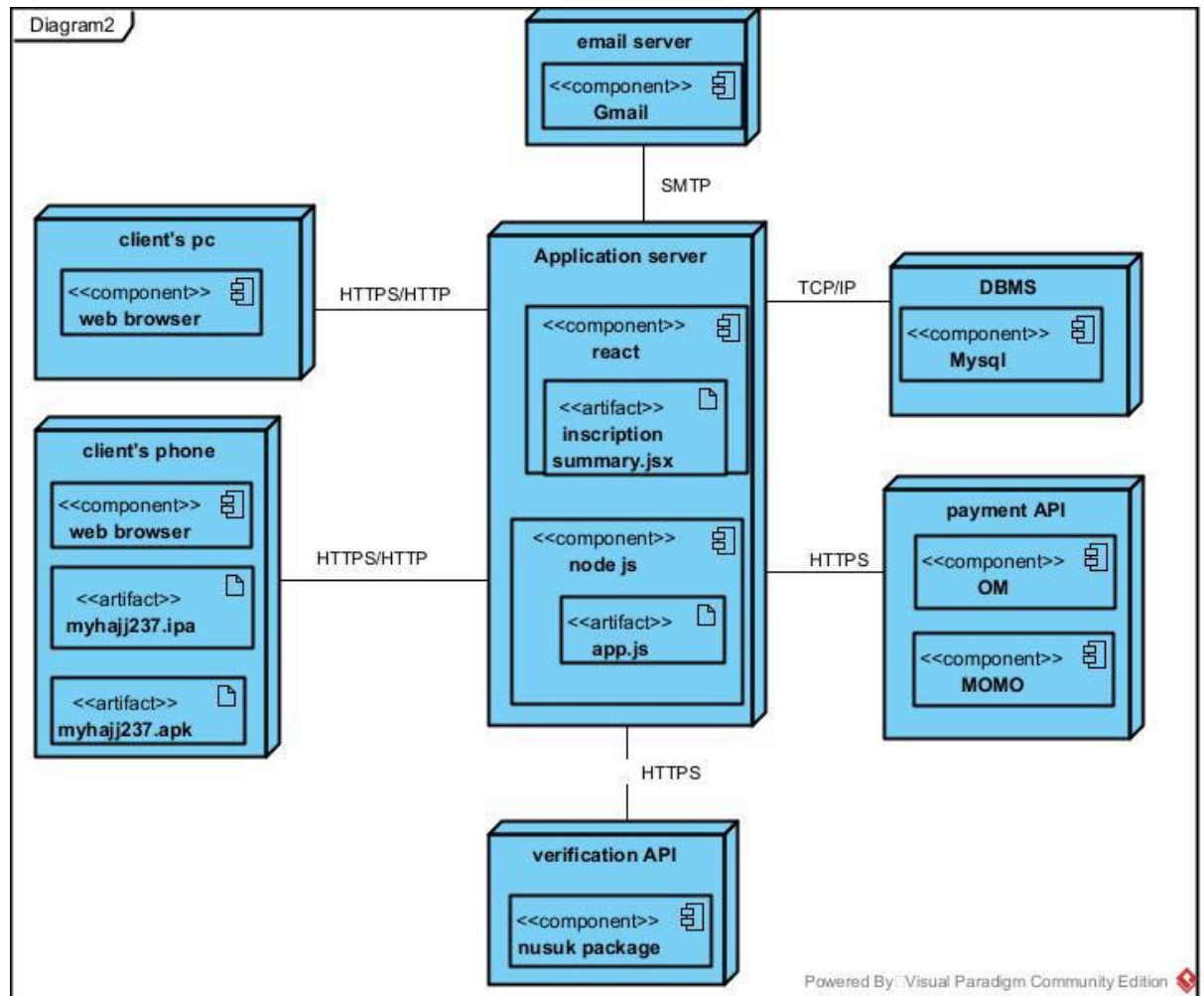


Figure 27: System package diagram

CONCLUSION

In the conception phase we set the stage for a successful project through the creation of key diagrams such as class diagrams, state machine diagrams, and package diagrams. This phase facilitates a shared understanding among stakeholders, identifies and mitigates potential risks and issues, and establishes a robust architectural foundation for implementation. We finished this phase by looking at interaction between the various aspects and actors of our system. The next phase of our report is the realization phase where we will look at aspects concerning the implementation of our system.

BOOK V : REALISATION PHASE

Preamble

In this phase we will to straight forward in the implementation of our solution, we will base ourselves on the analysis and conception phases and also present the component and deployment diagrams.

Content

INTRODUCTION

- I. ARCHITECTURE
 - 1. PHYSICAL ARCHITECTURE
 - 2. LOGICAL ARCHITECTURE
- II. DEPLOYMENT DIAGRAM
- III. COMPONENT DIAGRAM

INTRODUCTION

In the realization phase, we will talk about the architecture we use and some diagrams related to the physical aspect of the system like libraries, documents, as well as the physical topology of the components of the system when the software is been deployed.

PHYSICAL ARCHITECTURE

The design of the DBMS depends on its architecture. An n-tier architecture partitions on the whole system into related but separated n modules, which can be independently modified, altered, changed or replace. A large amount of data on web servers, personal computers (pc) and others are link with networks with the help of basic client or server architecture. Within the scope of our project, we made use of the n-tier architecture This architecture separates it tiers from each other based upon the user and the manipulated data in the database. Each layer has a well-defined communication interface, and the evolution of the layer is independent of the other. The n-tier of our system is made up of:

The client tier, which is also known as our presentation interphase.

Application Tier, which represents our webserver. The data tier, which represents our DBMS server.

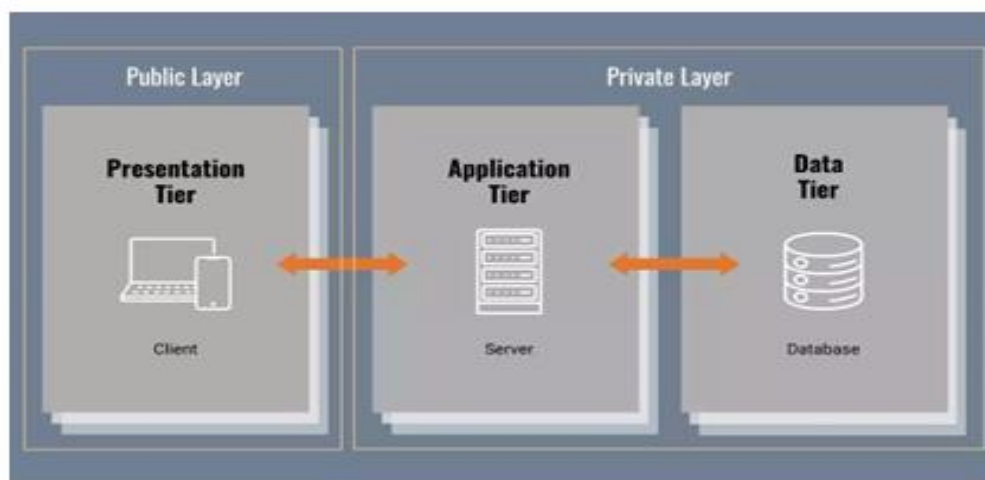


Figure 28: Physical architecture structure

LOGICAL ARCHITECTURE

The Model-View-Controller (MVC) framework is an architectural/design pattern that separates an application into three main logical components Model, View, and Controller. Each architectural component is built to handle specific development aspects of an application: Here's what each of those components mean

- Model: The backend that contains all the data logic
- View: The frontend or graphical user interface (GUI)
- Controller: The brains of the application that controls how data is displayed

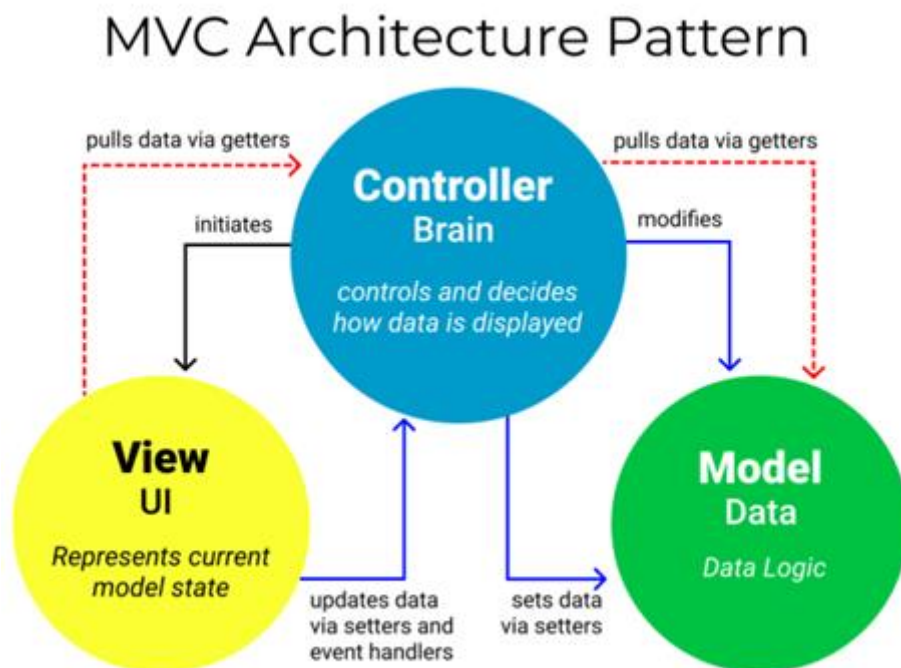


Figure 29: MVC architecture pattern

II. DEPLOYMENT DIAGRAM

Definition

Deployment diagram is a structural diagram used to visualize the topology of the physical components of a system, where the software is deployed. They consist of nodes and their relationship. It is related to the component diagram because the components are deployed using the deployment diagram. A deployment diagram consists of nodes (physical hardware used to deploy the application.).

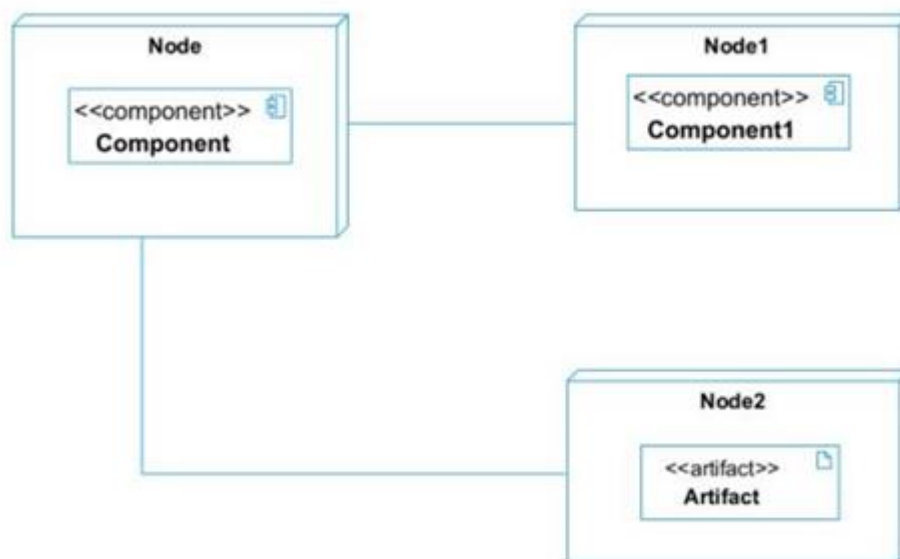
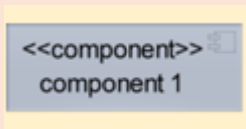
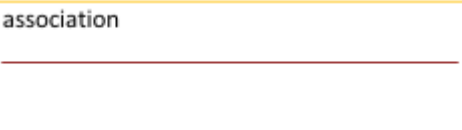


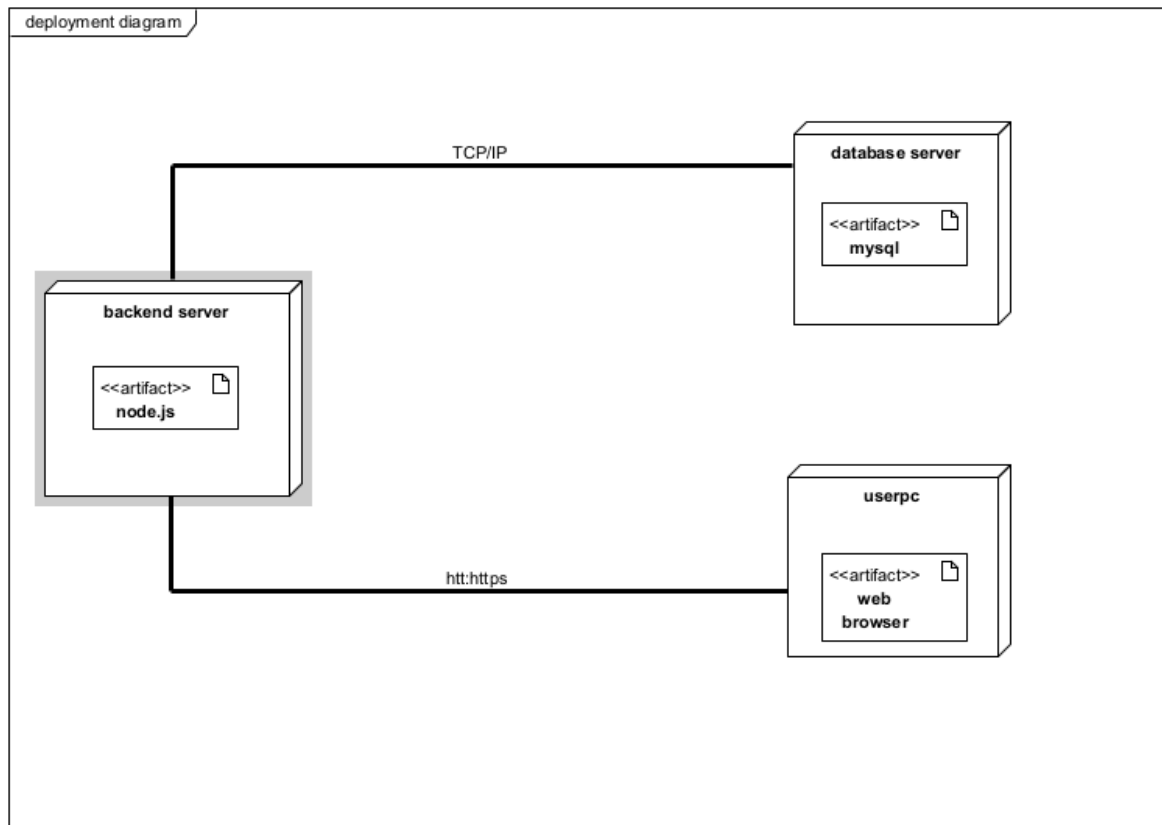
Figure 30:formalism of a deployment diagram

Components of a deployment diagram

Table 18 components of a deployment diagram

elements	representation	description
Node		It is a hardware used to deploy the application
Artifact		An artifact is a major product, which is produced or used during the development of a software e.g diagrams
Component		It represents a modular part of a system that encapsulates its content and whose manifestation is environment
Association		An association helps to connect two nodes together which permits them to communicate

.



Activ

Figure 31: System deployment diagram

III. COMPONENT DIAGRAM

Definition

Component diagrams are used to model the physical aspect of a system. Now the question is what are this physical aspect? They are elements such as Executables, libraries, files, document etc. which resides in a node. The component diagram does not describe the functionality of the system, but it describes the components used to make those functionalities.

Formalism

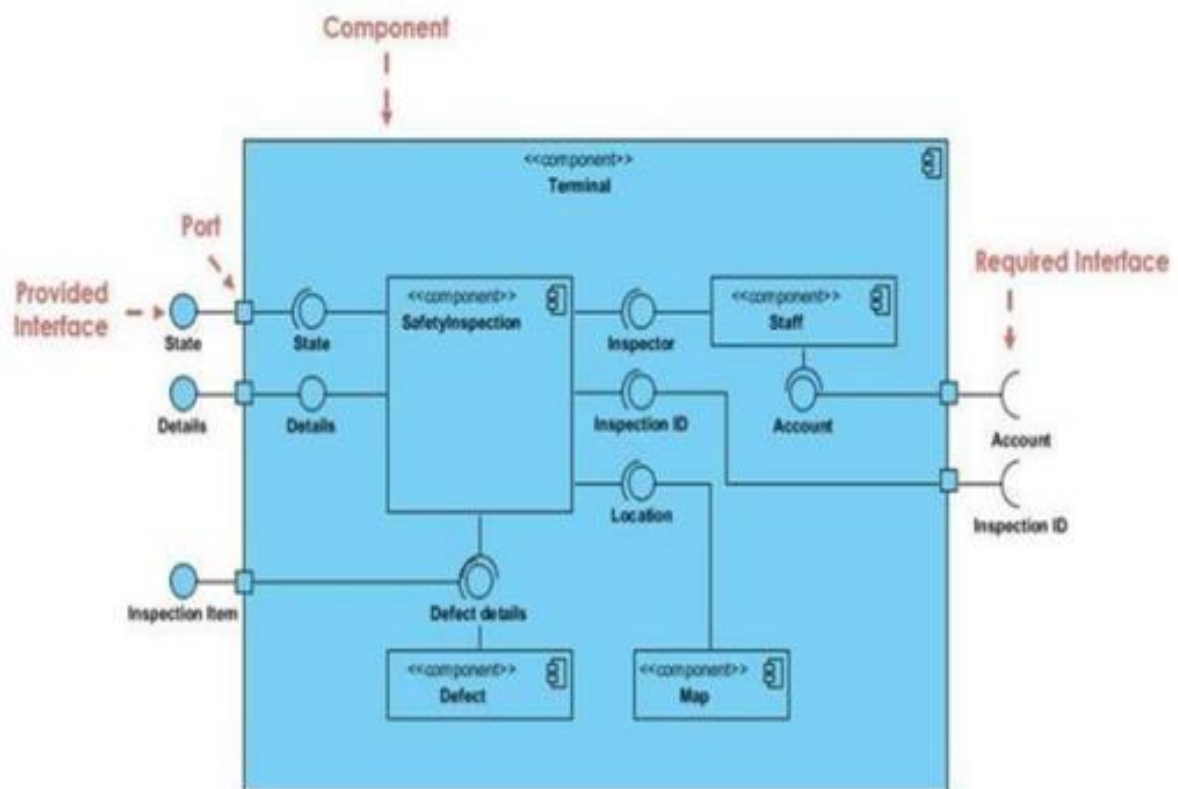
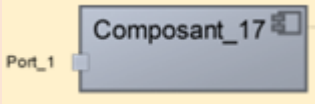
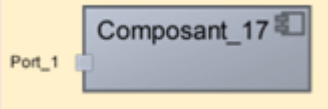


Figure 32: Component diagram formalism

Table 19 components of component diagram

Name	Representation	Description
Component		A component is an abstract logical unit block of a system
Dependency		It is a directed relationship which is used to show that some components are dependent on others for their correct function
Required interface		It is a straight line from the component box with an attached half cycle representing interfaces where a component requires information in order to perform its own function
Provide interface		It is a straight line from the component produces information used by required interfaces
Port		A port is used when the components delegate the interface to an internal class

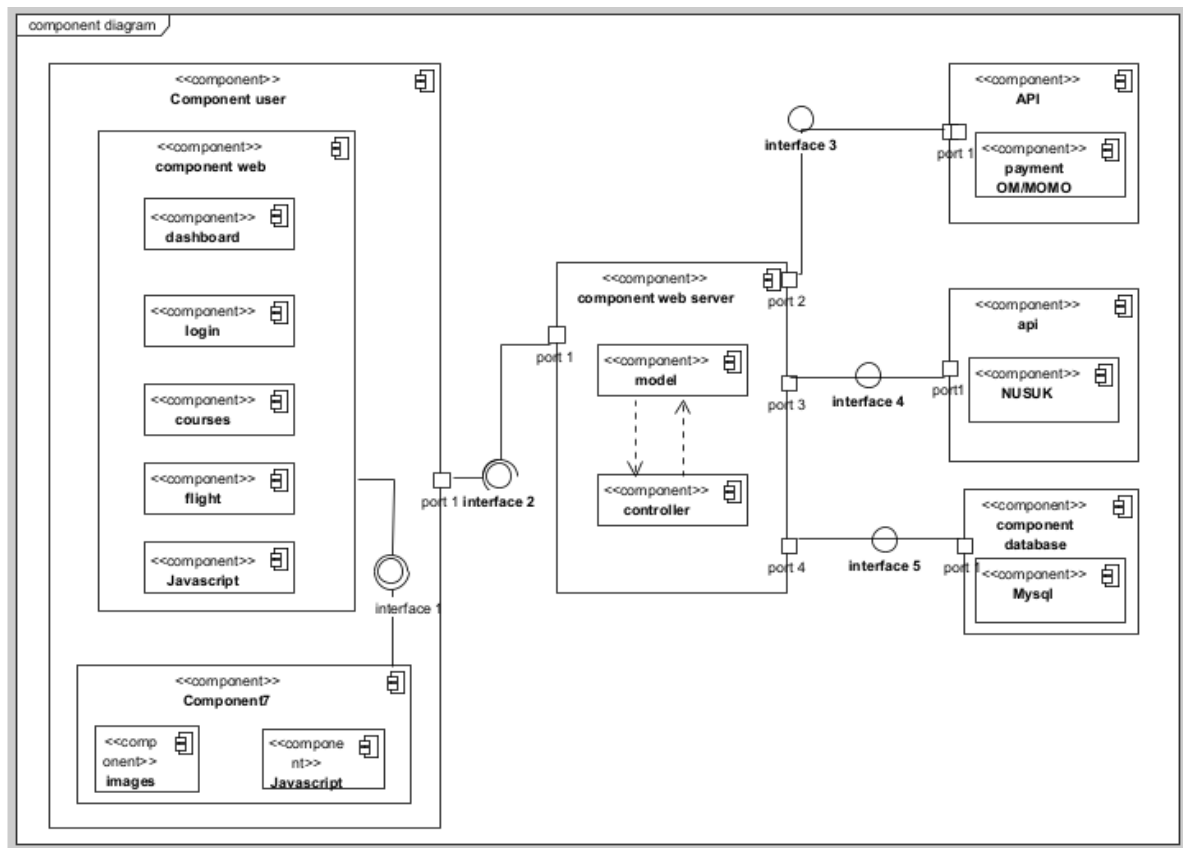


Figure 33:component diagram

Table 20:choice of tools

SOFTWARE	VERSION	ROLE	IMAGE
OS WINDOWS 10	10.0.19045	The operating system required to run the software is the one on which we have worked.	
Visual studio code	1.70.0.8	Text editor that permits code writing and debugging line by line	
GitHub	3.0.6	It is a website and cloud service that helps developers store and manage their code.	
Framework React And React Native	0.69.0	React JS is an English framework built on JavaScript for creating responsive web applications.	
Visual paradigm community	17.2	a professional visual modeling instrument used for building from scratch, creating, and editing complex diagrams for various needs, including internal processes, architecture frameworks, and business implementations.	
MySQL	9.3.0	Is a popular and widely used database management system that supports SQL and various operating systems	
Tailwind CSS	3.4	Tailwind CSS 3.4 provides a comprehensive and customizable colour palette designed for a wide range of design needs. The default palette includes 11 shades for each colour, ranging from 50 (lightest) to 950 (darkest). These colours can be applied to various utilities such as background, text, borders, shadows, and more.	

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Table 21:programming language and framework

Choice of programming language and framework

Image	Name	Usage
	Framework React and React Native	Front-end development
	Node Js	Back-end development
	CSS	Style-sheet for the frontend
	Express Js	Creating a virtual server

CONCLUSION

In our realization phase, we implemented our application. In other to accomplish this phase, we made use of our analysis and conception phase. We also drew the deployment and component diagrams which depict the structure of our system in terms of modules, files, assets, how the different elements interact with each other. We will move to the test of functionalities phase, where we will examine the different modules present in our app and how beneficial they are to its different users.

BOOK VI : TEST OF FUNCTIONALITIES

INTRODUCTION

The test of functionalities phase helps us to know more about the solution we are building. It provides the different functionalities or modules found in our application and how they are beneficial to the users. Hence, in this phase we are going to explore the different functionalities present in our application.

APPLICATION FUNCTIONALITIES

Authentication

This functionality enables the users to have access to their workspace or dashboard. In case he/she does not have an account he/she will register

Make payment

This functionality enables the pilgrims to make payments and get enrolled for their trip.

Manage course

Here the pilgrim guide is the one performing the action since he adds, deletes, and update courses.

Manage account

The admin is the one concerned here he can delete accounts or even coordinate registrations

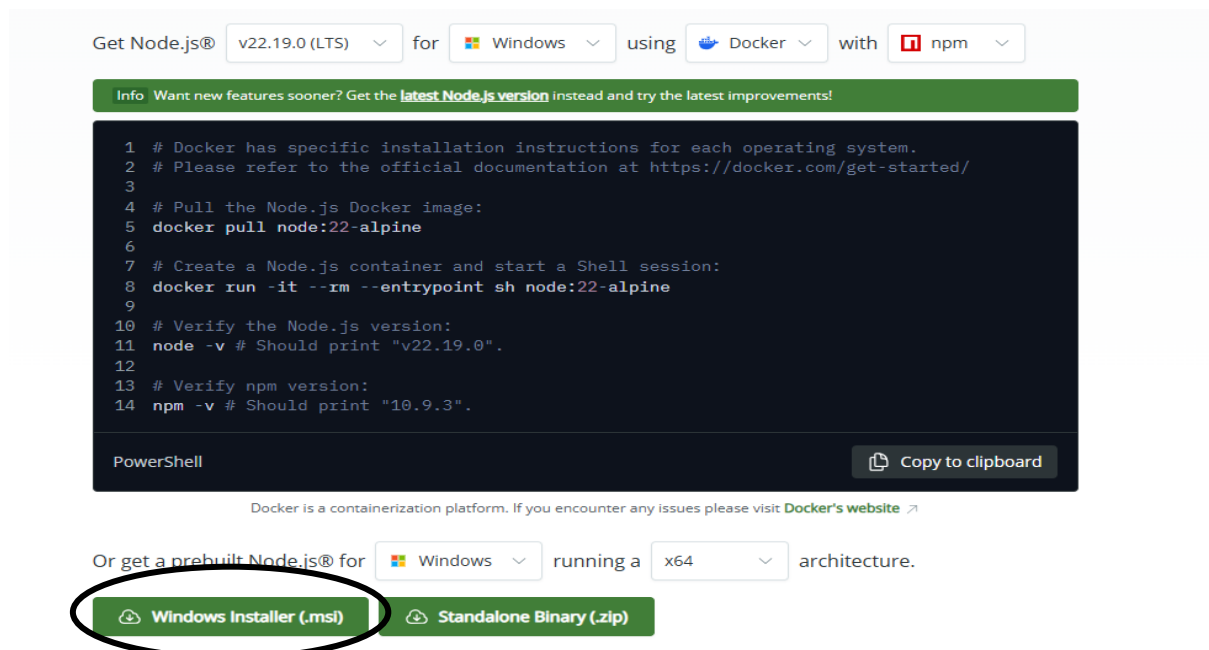
CONCLUSION

In the test of functionalities phase we have explained the different functionalities of our application and the benefits to the various users (student, schools, admin and orientators). This phase is essential in the understanding of the application. We will move to the last phase which is the installation and user guide where we will elaborate on how to install and use our application.

BOOK VII: INSTALLATION AND USER GUIDE

INSTALLATION OF THE APPLICATION

Step 1 : Download and install Node.js in your computer



Get Node.js® v22.19.0 (LTS) for Windows using Docker with npm

Info Want new features sooner? Get the latest Node.js version instead and try the latest improvements!

```

1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:22-alpine
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:22-alpine
9
10 # Verify the Node.js version:
11 node -v # Should print "v22.19.0".
12
13 # Verify npm version:
14 npm -v # Should print "10.9.3".

```

PowerShell Copy to clipboard

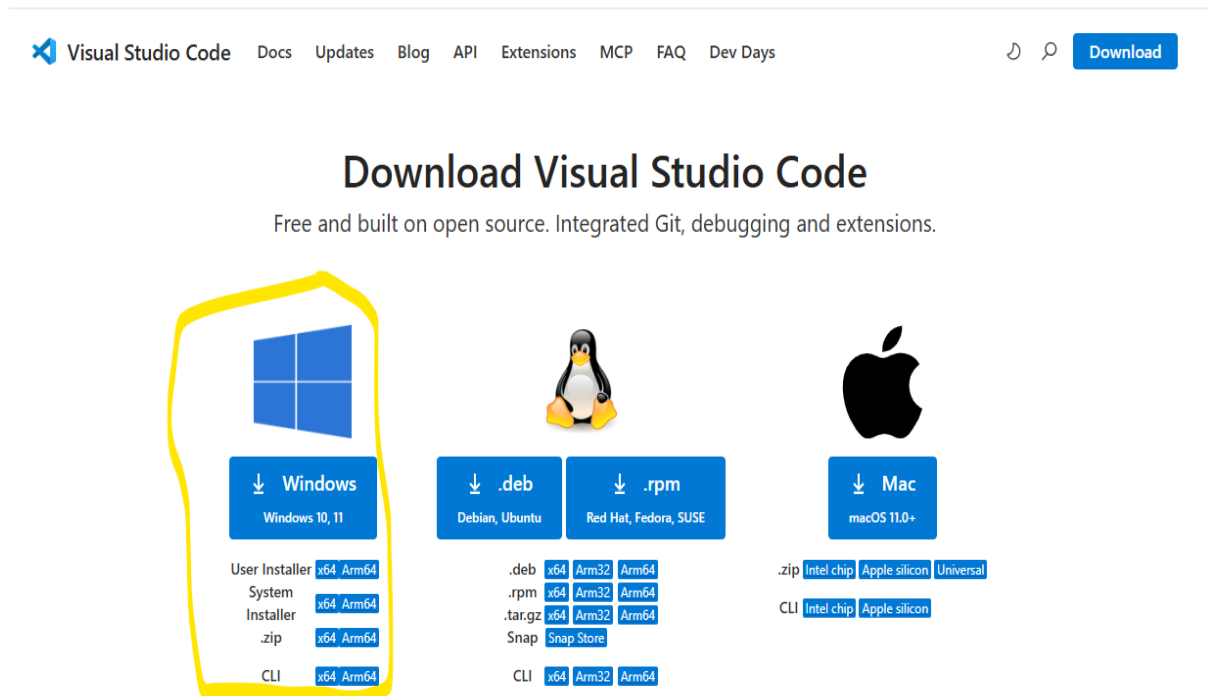
Docker is a containerization platform. If you encounter any issues please visit [Docker's website](#)

Or get a prebuilt Node.js® for Windows running a x64 architecture.

Windows Installer (.msi) Standalone Binary (.zip)

Figure 34: Download Node.Js

Step 2 : Download and install Visual Studio Code



Visual Studio Code Docs Updates Blog API Extensions MCP FAQ Dev Days

Download

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.

Windows Windows 10, 11

User Installer x64 Arm64
System Installer x64 Arm64
.zip x64 Arm64
CLI x64 Arm64

.deb x64 Arm32 Arm64
.rpm x64 Arm32 Arm64
.tar.gz x64 Arm32 Arm64
Snap Snap Store
CLI x64 Arm32 Arm64

Mac macOS 11.0+

.zip Intel chip Apple silicon Universal
CLI Intel chip Apple silicon

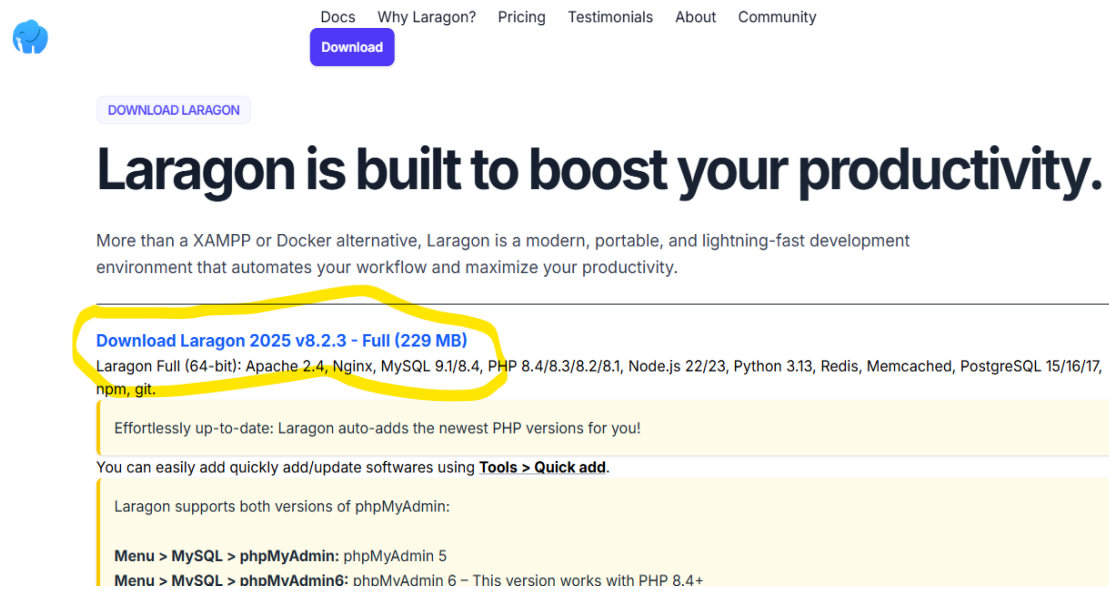
Figure 35: Download Visual Studio Code

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Step 3: Download and install Laragon

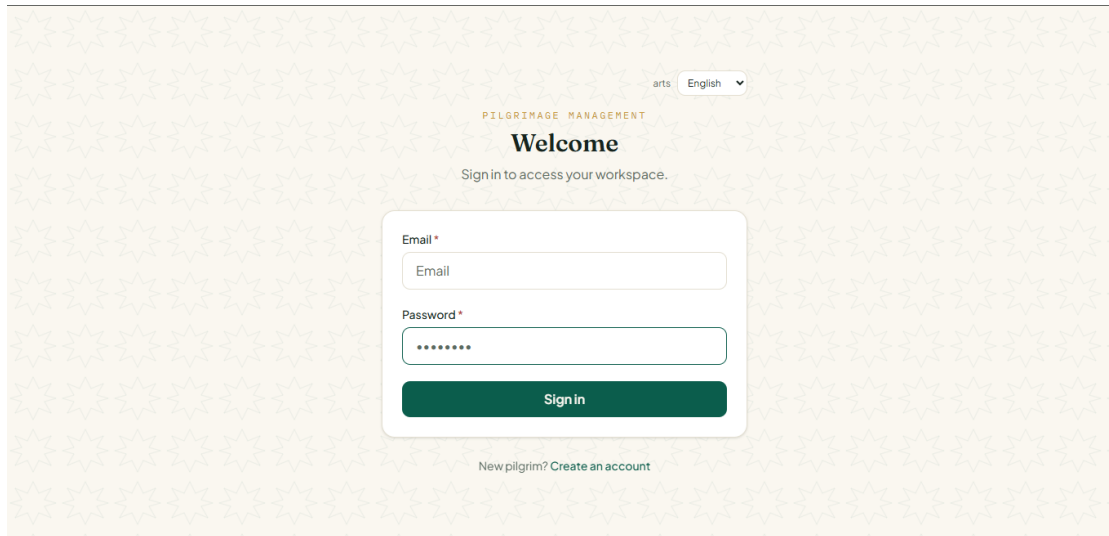


The screenshot shows the Laragon website homepage. At the top, there is a navigation bar with links: Docs, Why Laragon?, Pricing, Testimonials, About, and Community. A blue 'Download' button is prominently displayed. Below the navigation bar, a light blue button reads 'DOWNLOAD LARAGON'. The main heading is 'Laragon is built to boost your productivity.' followed by a subtext: 'More than a XAMPP or Docker alternative, Laragon is a modern, portable, and lightning-fast development environment that automates your workflow and maximize your productivity.' A yellow circle highlights the download link: 'Download Laragon 2025 v8.2.3 - Full (229 MB)'. Below this, a list of included software is shown: 'Laragon Full (64-bit): Apache 2.4, Nginx, MySQL 9.1/8.4, PHP 8.4/8.3/8.2/8.1, Node.js 22/23, Python 3.13, Redis, Memcached, PostgreSQL 15/16/17, npm, git.' Three yellow callout boxes provide additional information: 'Effortlessly up-to-date: Laragon auto-adds the newest PHP versions for you!', 'You can easily add quickly add/update softwares using **Tools > Quick add.**', and 'Laragon supports both versions of phpMyAdmin:
Menu > MySQL > phpMyAdmin: phpMyAdmin 5
Menu > MySQL > phpMyAdmin6: phpMyAdmin 6 – This version works with PHP 8.4+'.

Figure 36: Download Laragon

App screenshots

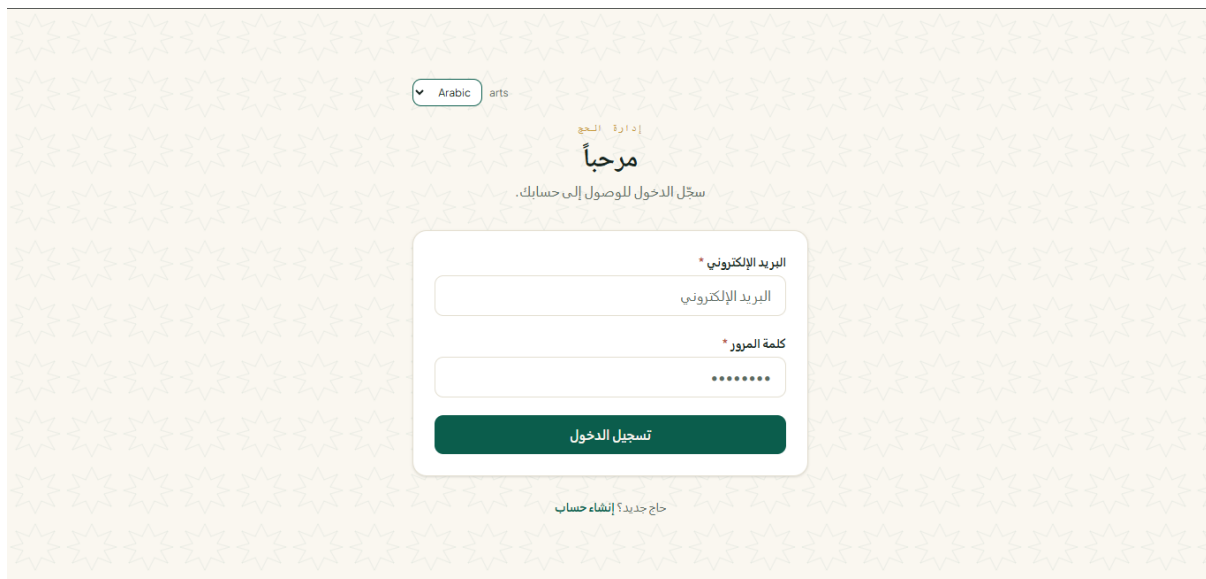
Login page



The screenshot shows the English version of the login page. At the top right, there is a language dropdown menu set to 'English'. Below it, the text 'PILGRIMAGE MANAGEMENT' is displayed. The main heading is 'Welcome', followed by the instruction 'Sign in to access your workspace.' The login form consists of two input fields: 'Email *' and 'Password *'. The 'Email' field contains the placeholder text 'Email', and the 'Password' field contains a series of dots. Below the password field is a green 'Sign in' button. At the bottom, there is a link that says 'New pilgrim? Create an account'.

Figure 37: Login page

Login page in Arabic



The screenshot shows the Arabic version of the login page. At the top left, there is a language dropdown menu set to 'Arabic'. Below it, the text 'إدارة الحج' (Hajj Management) is displayed. The main heading is 'مرحباً' (Welcome), followed by the instruction 'سجل الدخول للوصول إلى حسابك.' (Sign in to access your account). The login form consists of two input fields: 'البريد الإلكتروني *' (Email) and 'كلمة المرور *' (Password). The 'Email' field contains the placeholder text 'البريد الإلكتروني', and the 'Password' field contains a series of dots. Below the password field is a green button labeled 'تسجيل الدخول' (Sign in). At the bottom, there is a link that says 'حاج جديد؟ إنشاء حساب' (New pilgrim? Create an account).

Figure 38: Registration page

Dashboard

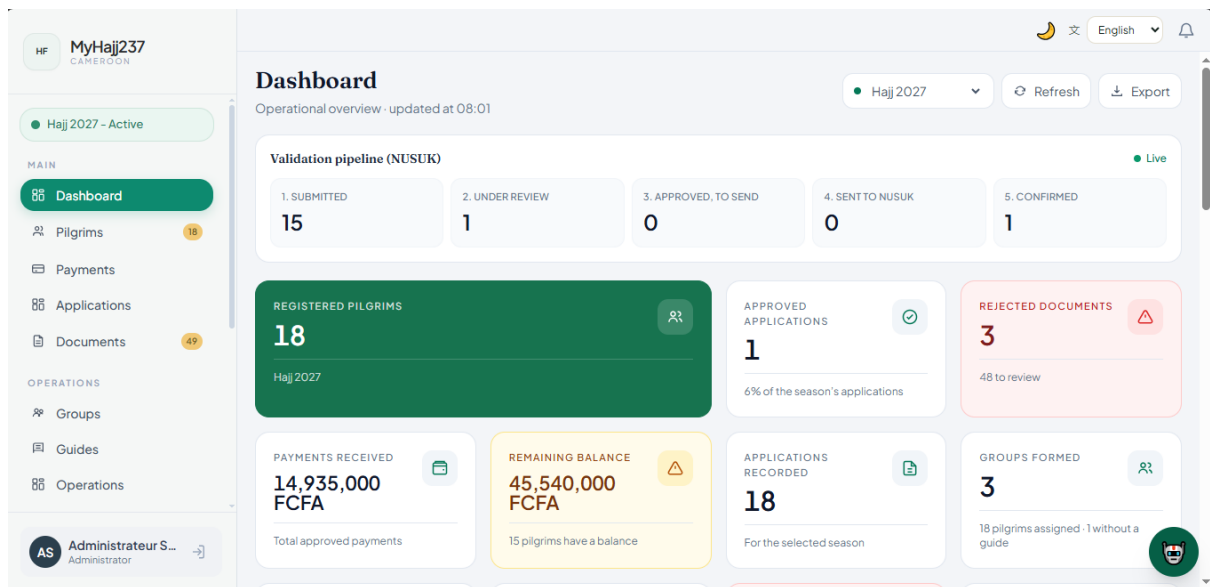
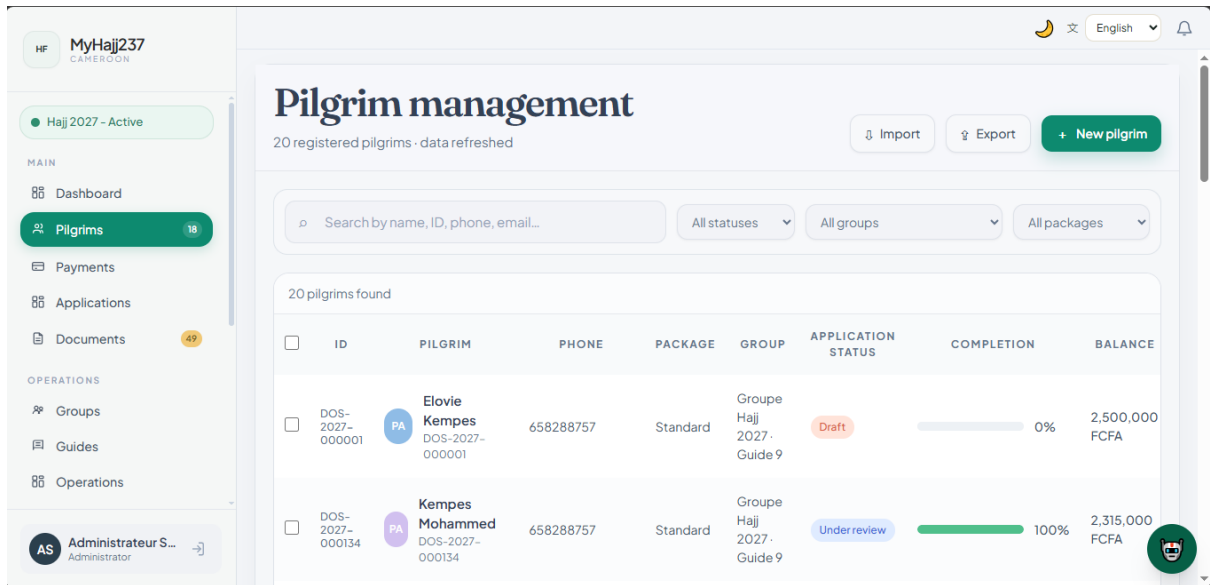
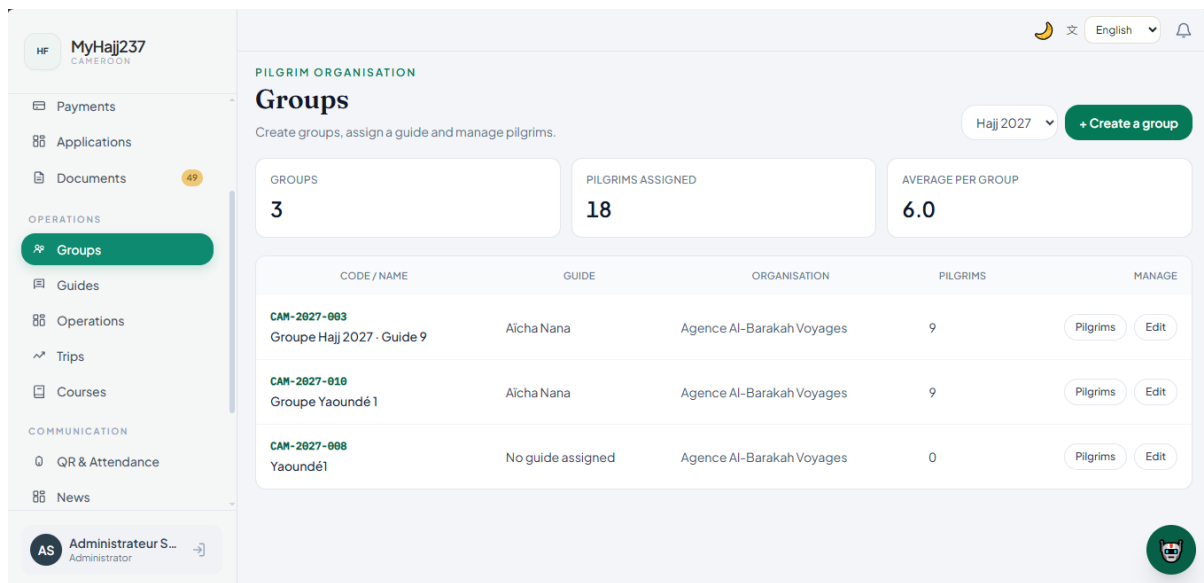


Figure 39: Dashboard



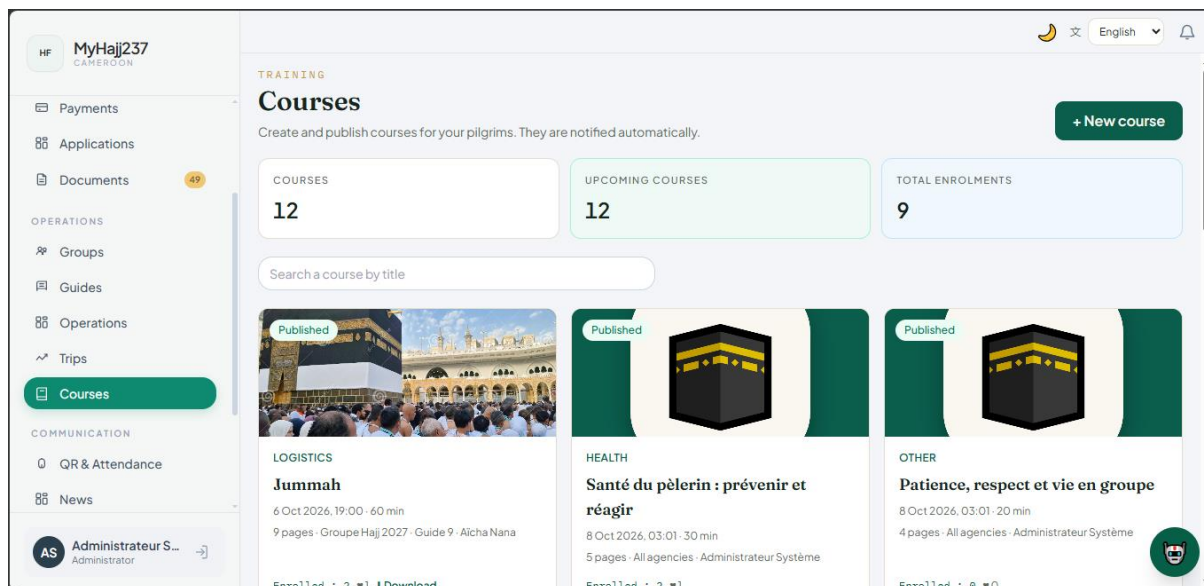
The Pilgrim management page shows 20 registered pilgrims. The table below displays the first two entries:

ID	PILGRIM	PHONE	PACKAGE	GROUP	APPLICATION STATUS	COMPLETION	BALANCE
DOS-2027-000001	Elovie Kempes DOS-2027-000001	658288757	Standard	Groupe Hajj 2027 - Guide 9	Draft	0%	2,500,000 FCFA
DOS-2027-000134	Kempes Mohammed DOS-2027-000134	658288757	Standard	Groupe Hajj 2027 - Guide 9	Under review	100%	2,315,000 FCFA



CODE / NAME	GUIDE	ORGANISATION	PILGRIMS	MANAGE
CAM-2027-003 Groupe Hajj 2027 - Guide 9	Aicha Nana	Agence Al-Barakah Voyages	9	Pilgrims Edit
CAM-2027-010 Groupe Yaoundé 1	Aicha Nana	Agence Al-Barakah Voyages	9	Pilgrims Edit
CAM-2027-008 Yaoundé1	No guide assigned	Agence Al-Barakah Voyages	0	Pilgrims Edit

Manage courses



COURSE	DATE	DURATION	ENROLLED
LOGISTICS Jummah 6 Oct 2026, 19:00 - 60 min 9 pages - Groupe Hajj 2027 - Guide 9 - Aicha Nana	6 Oct 2026, 19:00 - 60 min	9 pages - Groupe Hajj 2027 - Guide 9 - Aicha Nana	Enrolled : 2
HEALTH Santé du pèlerin : prévenir et réagir 8 Oct 2026, 03:01 - 30 min 5 pages - All agencies - Administrateur Système	8 Oct 2026, 03:01 - 30 min	5 pages - All agencies - Administrateur Système	Enrolled : 2
OTHER Patience, respect et vie en groupe 8 Oct 2026, 03:01 - 20 min 4 pages - All agencies - Administrateur Système	8 Oct 2026, 03:01 - 20 min	4 pages - All agencies - Administrateur Système	Enrolled : 0

Figure 40: Manage courses

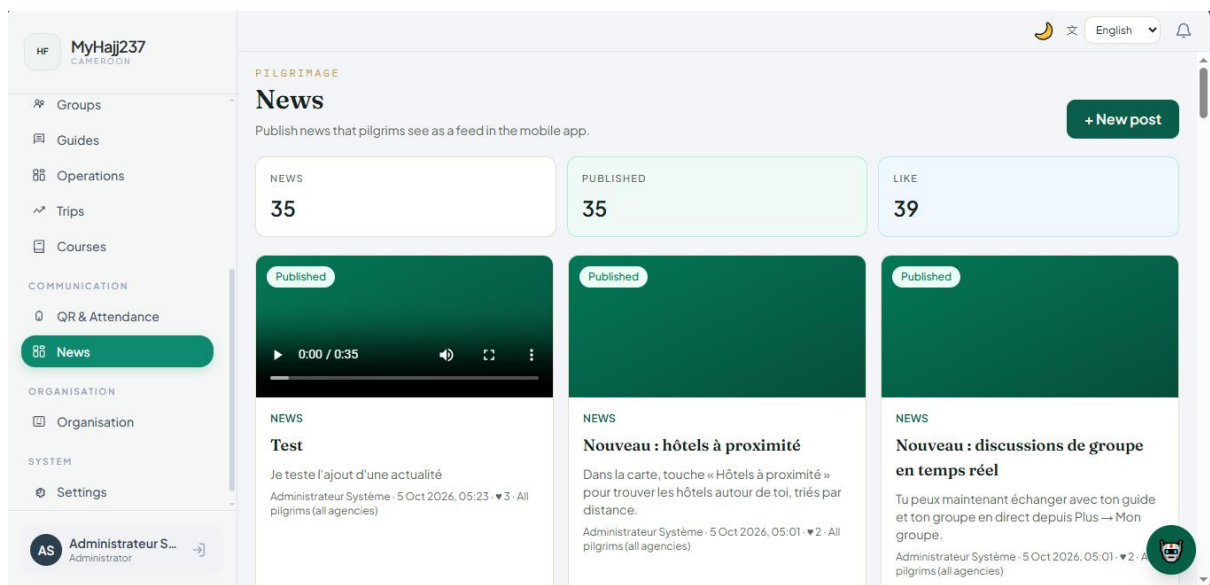


Figure 41: Manage news

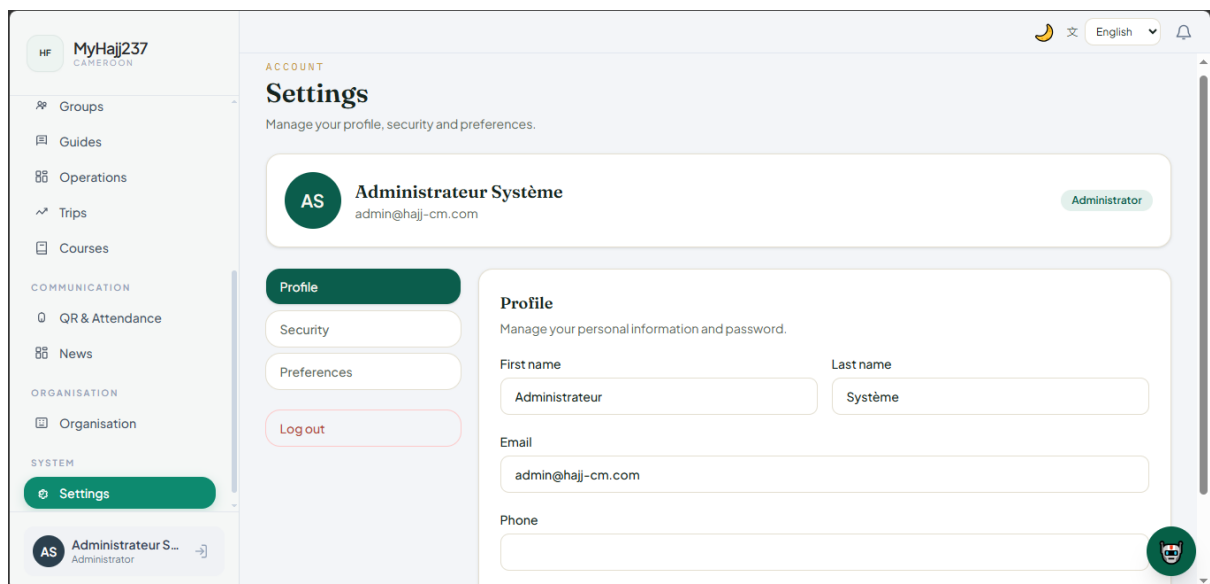


Figure 42: Manage profile

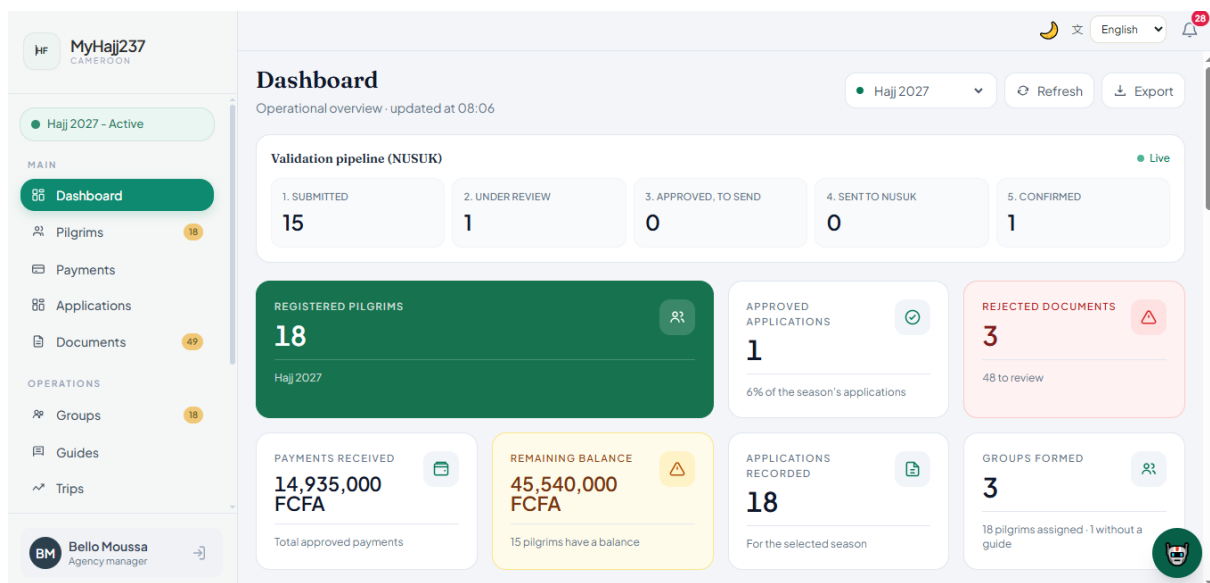


Figure 44: Agency dashboard

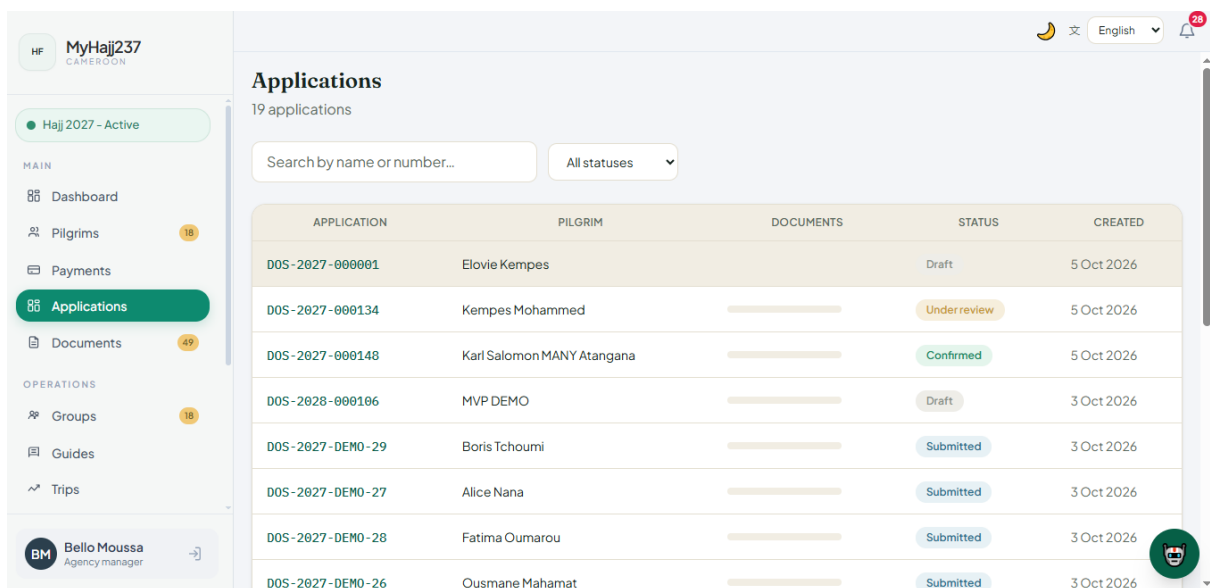


Figure 43: Manage application

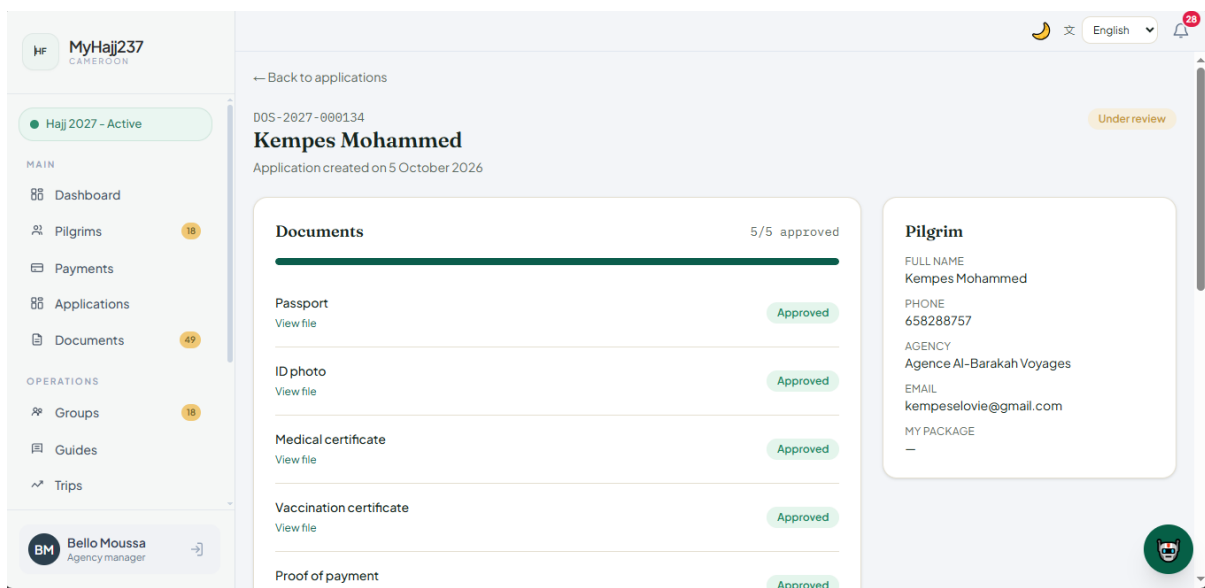


Figure 45: Pilgrim application

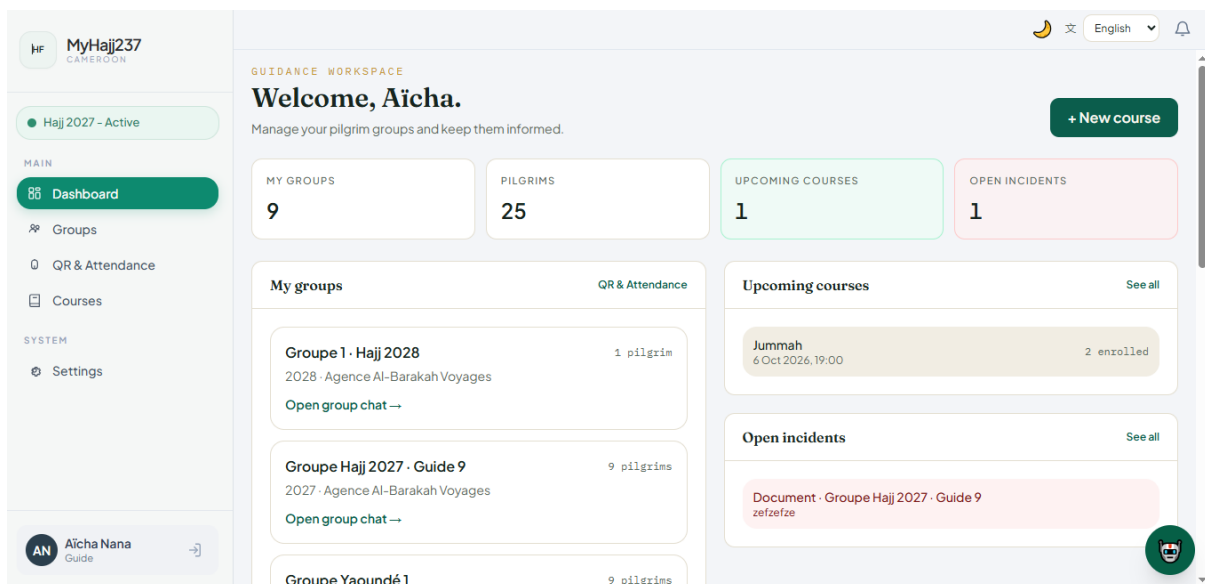


Figure 46: Guide dashboard

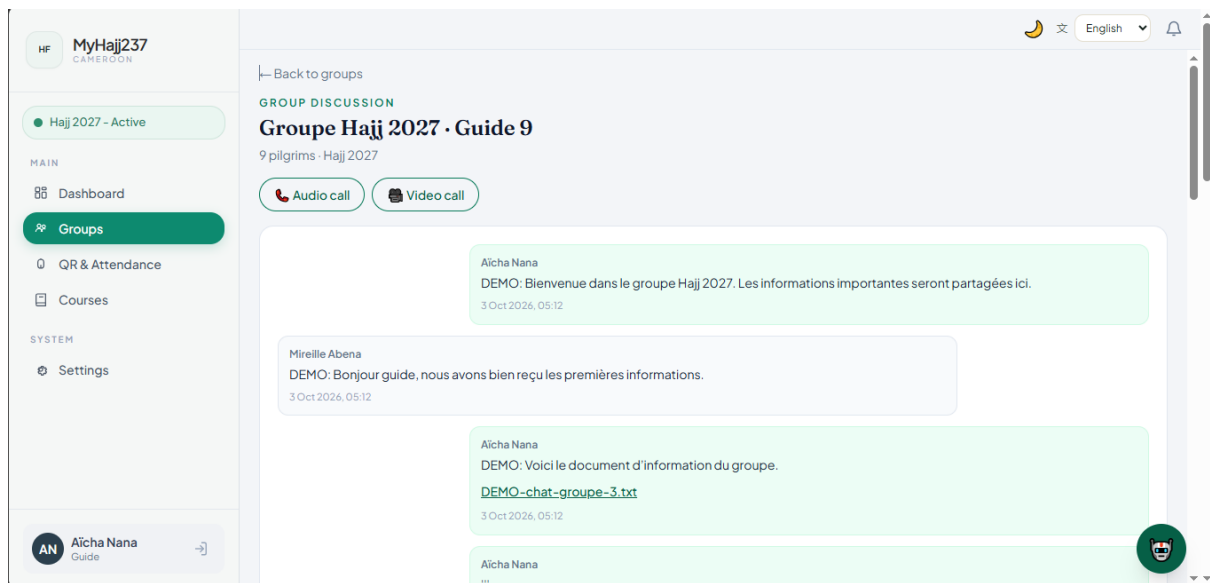


Figure 47: Group chat

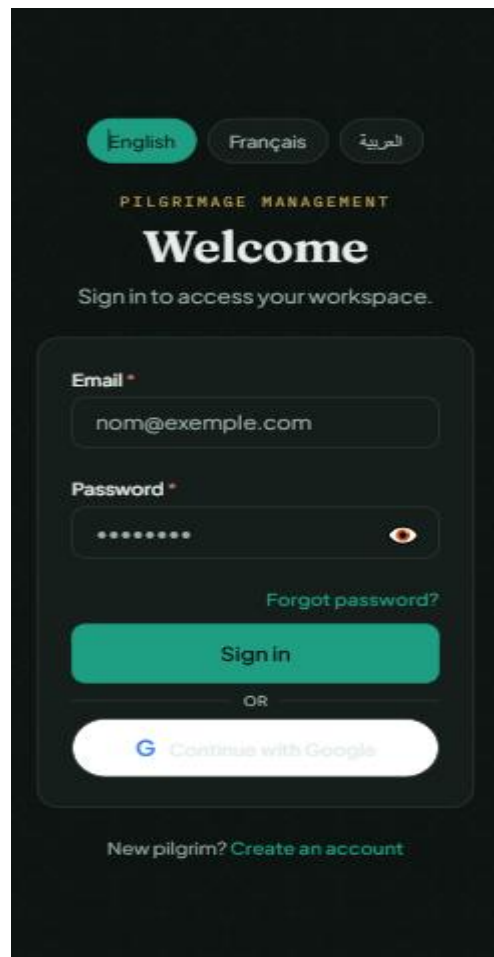


Figure 48: Mobile login screen

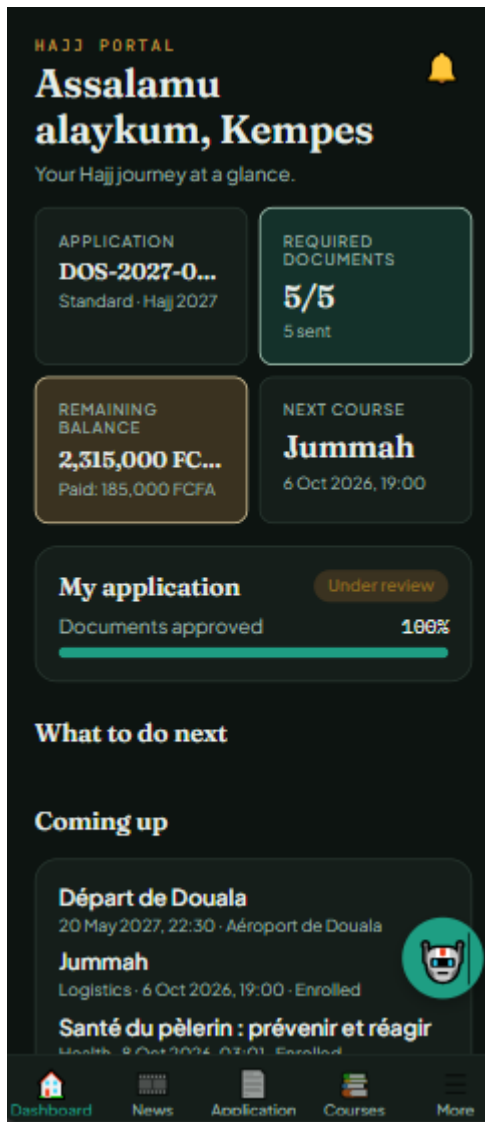


Figure 50: Pilgrim dashboard



Figure 49: News menu

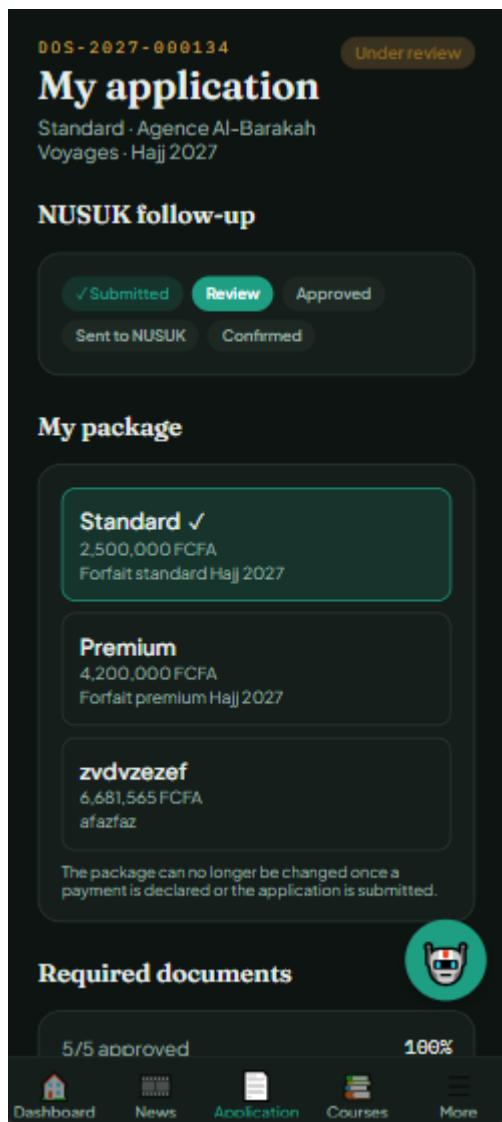


Figure 52: Pilgrim's application



Figure 51: View courses

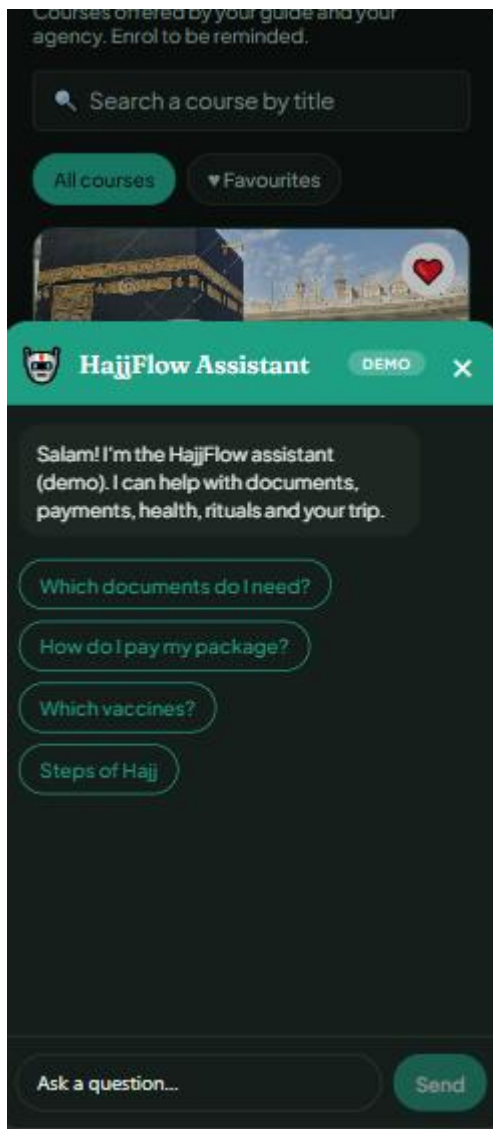


Figure 54: Our AI Assistant

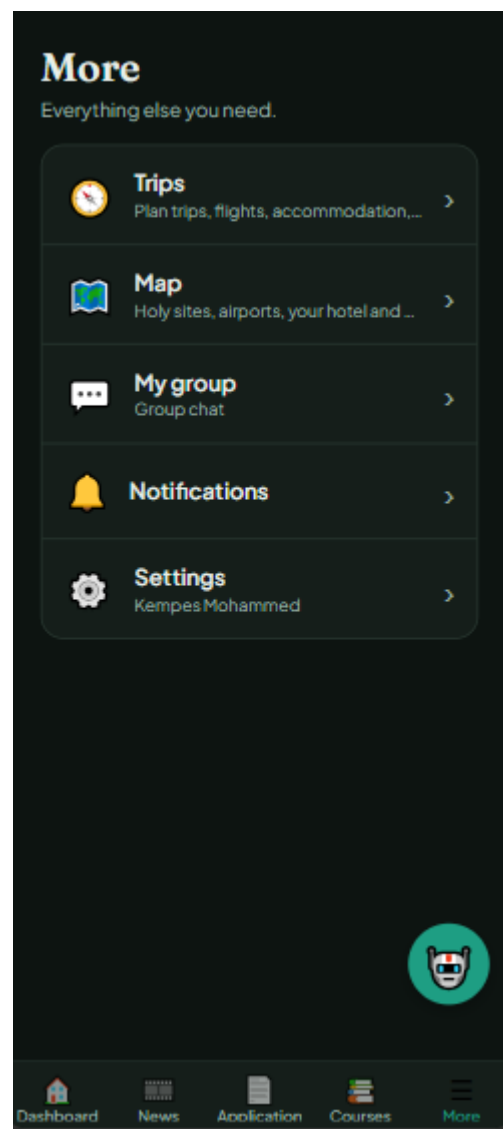


Figure 53: Others

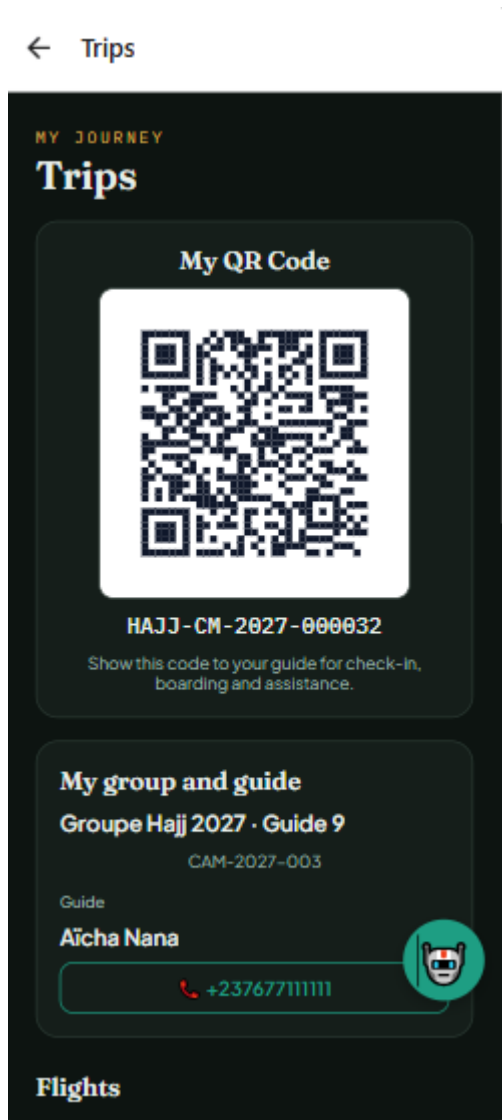


Figure 56: Pilgrim's Trips

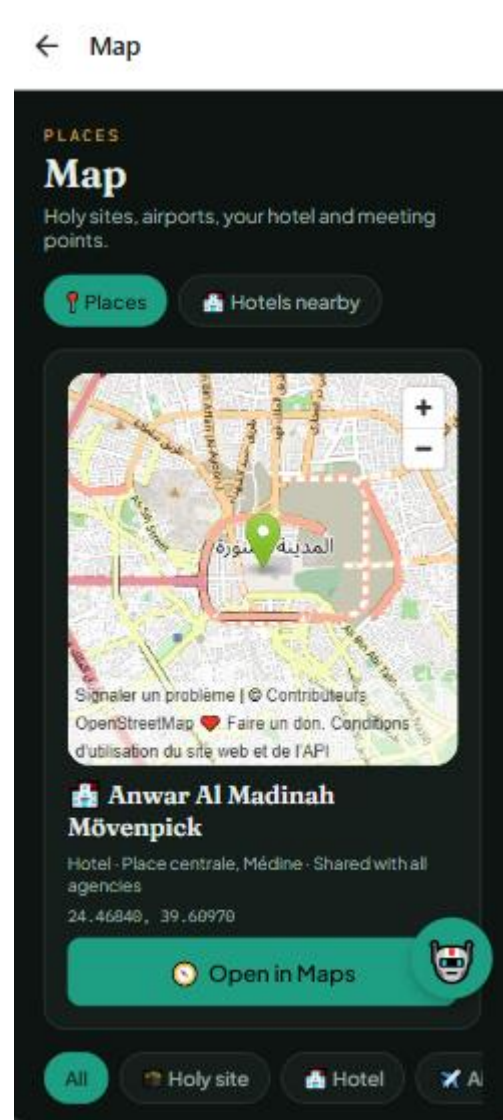


Figure 55: The map view for localisation



**DESIGN AND IMPLEMENTATION OF A CROSS-PLATFORM
INTEGRATED MANAGEMENT APPLICATION FOR HAJJ IN CAMEROON**



Written by **HAWA ADAMA MAMOUDOU,**

Level III Student Software Engineering at AICS CAMEROON PAUL

BIYA TECHNOLOGICAL CENTER OF EXCELLENCE **Academic year 2025-2026**

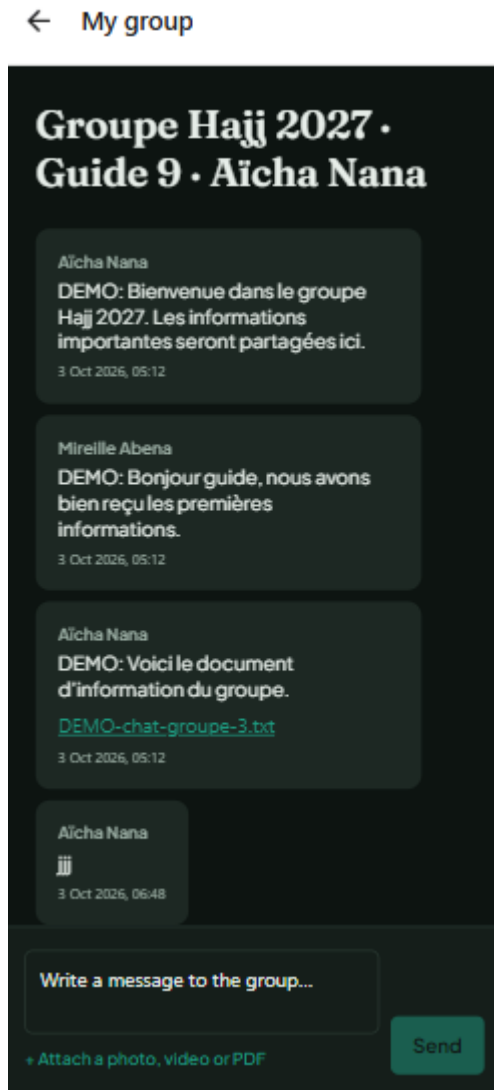


Figure 58: Group chat

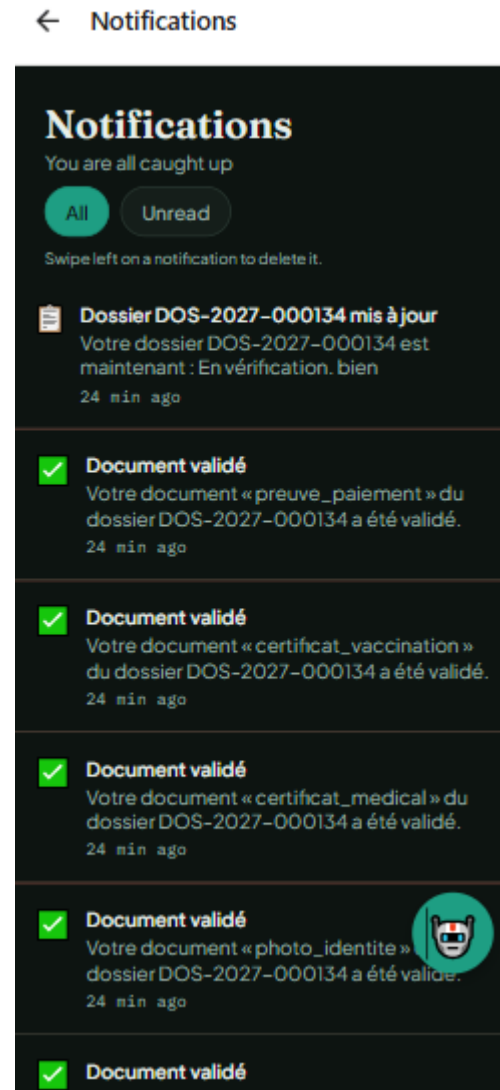


Figure 57: Notifications center

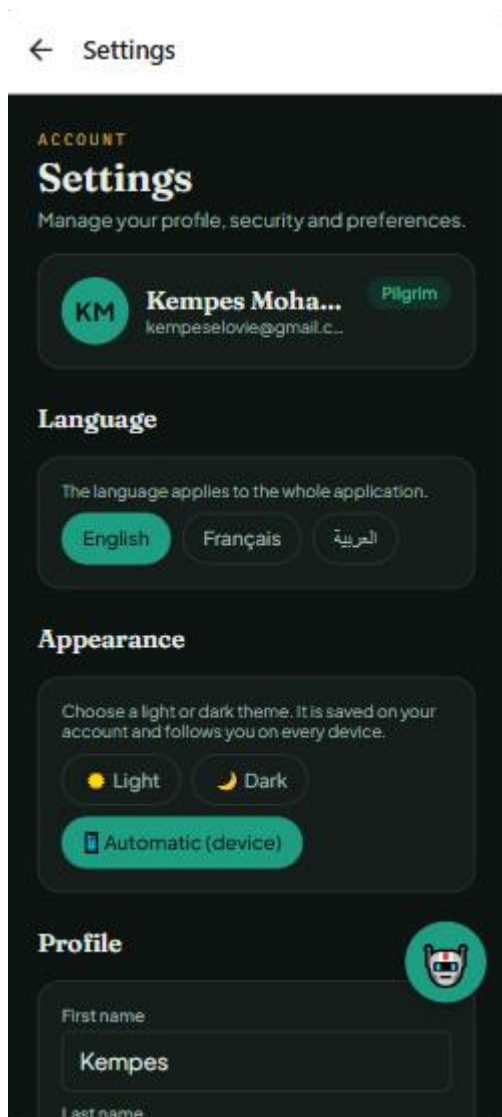


Figure 59: Pilgrim's settings

CONCLUSION

Having put in place the platform, it was not sufficient because we had to put in place a chat system or a video call conferencing which could enable pilgrim and pilgrim guide communicate. But we presented the different tools to be installed and how they are to be installed in order to enable users to be able to use application without any problem.

PERSPECTIVES

Considering the fact that this project can be improved to a higher extend, as perspectives we can bring some ameliorations such as:

- Implementation of a video call conference for meeting between pilgrims and pilgrims guides
- Putting in place an AI that will enable to ease the orientation session with the pilgrims
- Putting in place a Global positioning system (GPS) that will help pilgrims easily locate hotels

GENERAL CONCLUSION

The project to modernize the Hajj management system in Cameroon represents a silent revolution that transcends simple digitization to embody a radical transformation of public governance. By using blockchain technology and artificial intelligence to eliminate a corrupt system that embezzles 1 billion CFA francs annually, this project restores social justice by guaranteeing equitable access to the 10,000 official places for all Cameroonian Muslims. With a guaranteed return on investment in the second year and a 30% reduction in costs per pilgrim, this initiative positions Cameroon as a continental technology leader while reconciling modernity and spirituality. Beyond its immediate objectives, this exemplary model of total transparency and excellence in public service will create a transformative ripple effect across the entire Cameroonian administration. This technological revolution in the service of religious dignity redefines the possible standards of African public management, proving that innovation can completely eliminate corruption when it is put at the service of the most noble values. The pilgrimage to Mecca thus becomes once again an equal right for all believers, organized with excellence in a totally transparent system that honors the highest aspirations of the Cameroonian nation.



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